Deep Speech Processing

ASR module

Instructor
Dr. Mrinmoy Bhattacharjee
Assistant Professor, Dept. of CSE,
IIT Jammu

Syllabus

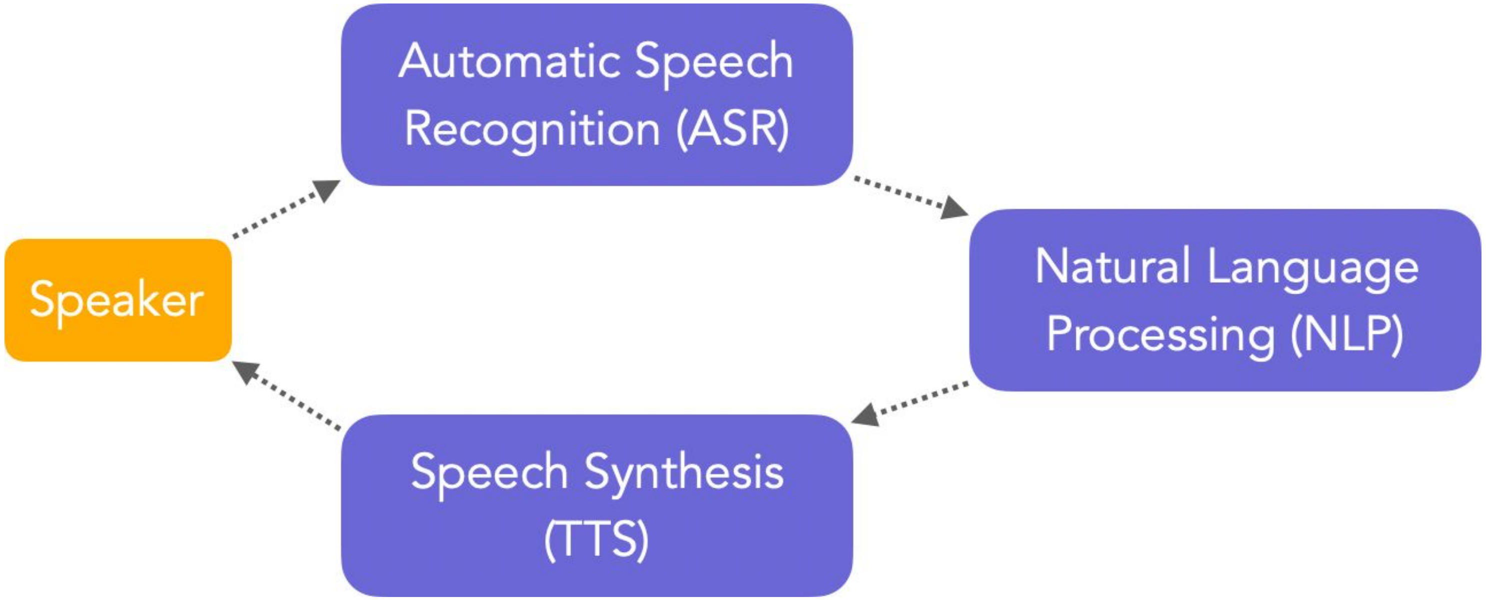
- Introduction and brief history
  - Types of ASR systems (Cascaded, End-to-End),
  - Evaluation metric - WER, CER
- Cascaded ASR system -
  - acoustic model (GMM-HMM, DNN-HMM, TDNN),
  - language models (n-gram, RNN),
  - WFST decoding, greedy decoding, beam search decoding
- E2E ASR systems
  - Encoder-Decoder,
  - CTC, RNN-T
  - ASR using LLMs
- Applications
  - Streaming vs. Non-Streaming ASR
  - Popular ASR systems - Whisper, wav2vec2.0;
  - Variants of Transformers;
  - Popular datasets,
  - Toolkits,
  - Open problems,
- Introduction to Mamba for ASR

What is speech recognition?

- Automatic Speech Recognition (ASR), or Speech-to-text (STT) is a field of study that aims to transform raw audio into a sequence of corresponding words.
- Speech-to-text transcription
  - Transform recorded audio into a sequence of words
  - Just the words, no meaning.... But we need to deal with acoustic ambiguity:

    “Recognise speech?” or “Wreck a nice beach?”

Classical pipeline in an ASR-powered application



Picture courtesy: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr

Comparison with other speech technologies

- **Speaker verification: Who spoke?**

Comparison with other speech technologies

- Speaker verification: Who spoke?
- **Speaker diarization: Who spoke when?**

Comparison with other speech technologies

- Speaker verification: Who spoke?
- Speaker diarization: Who spoke when?
- **Paralinguistic aspects: how did they say it? (timing, intonation, voice quality)**

Comparison with other speech technologies

- Speaker verification: Who spoke?
- Speaker diarization: Who spoke when?
- Paralinguistic aspects: how did they say it? (timing, intonation, voice quality)
- **Speech understanding: what does it mean?**

Comparison with other speech technologies

- Speaker verification: Who spoke?
- Speaker diarization: Who spoke when?
- Paralinguistic aspects: how did they say it? (timing, intonation, voice quality)
- Speech understanding: what does it mean?
- **Speech recognition: what did they say?** Nothing more, nothing less

Popular use-cases of speech recognition

- Voice assistants

Google Assistant | Apple Siri | Microsoft Cortana | Amazon Alexa

- Voice Search
- Automatic captioning on YouTube

Popular use-cases of speech recognition

- **Medical speech to text**: Amazon Transcribe Medical, Deepgram
- **In-vehicle infotainment systems**: Ford’s Sync, General Motor’s OnStar
- **Customer service**
- **Business applications**

Significance of ASR in the future

- **Enhancing human computer interaction**

Significance of ASR in the future

- Enhancing human computer interaction
- **Improved accessibility**

Significance of ASR in the future

- Enhancing human computer interaction
- Improved accessibility
- **Revolutionizing healthcare**

Significance of ASR in the future

- Enhancing human computer interaction
- Improved accessibility
- Revolutionizing healthcare
- **Education and learning**

Significance of ASR in the future

- Enhancing human computer interaction
- Improved accessibility
- Revolutionizing healthcare
- Education and learning
- **Automation and efficiency**

Significance of ASR in the future

- Enhancing human computer interaction
- Improved accessibility
- Revolutionizing healthcare
- Education and learning
- Automation and efficiency
- **Data and knowledge extraction**

Why is speech recognition difficult?

From a linguistic perspective

Speaker	Tuned for a particular speaker, or speaker-independent? Adaptation to speaker characteristics
Environment	Noise, competing speakers, channel conditions (microphone, phone line, room acoustics)
Style	Continuously spoken or isolated? Planned monologue or spontaneous conversation?
Vocabulary	Machine-directed commands, scientific language, colloquial expressions
Accent/dialect	Recognise the speech of all speakers who speak a particular language
Other paralinguistics	Emotional state, social class, etc.
Language spoken	Estimated 7,000 languages worldwide, most with limited training resources; code-switching; language change

From a machine learning perspective

- As a classification problem: **very high dimensional output space**
- As a sequence-to-sequence problem: **very long input sequences** (although limited re-ordering between acoustic and word sequences)
- Data is often noisy, with many “nuisance” factors of **variation in the data**
- Very **limited training data** available (in terms of words) compared to text-based NLP
    - Manual speech transcription is very expensive (10x real time)
- Hierarchical and compositional nature of speech production and comprehension makes it difficult to handle with a single model

[Appendix]

History of ASR

- Human interest in recognizing and synthesizing speech dates back hundreds of years (at least!)
  - First ASR system in mid 20th century
- Earliest examples:
  - A “digit recognizer” called **Audrey by Bell Labs** – 1952
    - Distinguished digits using “formants”
  - **IBM’s Shoebox** – recognized digits and arithmetic commands like “plus” and “total”
    - Passed the math problem to an adding machine to calculate and print results
  - Researchers at Japan and England built machines to recognize vowels and consonants

IBMs shoebox

The Pierce Freeze

Reprinted from THE JOURNAL OF THE ACOUSTICAL SOCIETY OF AMERICA, Vol. 46, No. 4, (Part 2), 1049–1051, October 1969

Whither Speech Recognition?

J.R. PIERCE

Bell Telephone Laboratories, Inc., Murray Hill, New Jersey 07971

Speech recognition has glamor. Funds have been available. Results have been less glamorous. "When we listen to a person speaking—much of what we think we—hear

computer to convince the listener that it was so deep as to be ... rovingian accent, ... understand what ... ral American. ... interesting game

Speech recognition has glamor. Funds have been available. Results have been less glamorous. "When we listen to a person speaking—much of what we think we—hear is supplied from our memory. [W. James, *Talks to Teachers on Psychology and to Students on Some of Life's Ideals* (Holt, New York, 1889), p. 159]. General-purpose speech recognition seems far away Special-purpose speech recognition is severely limited. It would seem appropriate for people to ask themselves why they are working in the field and what they can expect to accomplish.

There was optimism elsewhere

- In the early 1970s, the **U.S. DARPA** funded a 5-year program called Speech Understanding Research
  - This led to the creation of several new ASR systems
  - Carnegie Mellon University's Harpy was the most successful one
  - Could recognize just over 1000 words by 1976.
- IBM and AT&T's Bell Laboratories pushed the technology toward possible commercial applications
  - IBM prioritized speech transcription in the context of office correspondence
  - Bell was concerned with "command and control" scenarios
    - Precursors to the voice dialing and automated phone trees

80’s and beyond

- **Hidden Markov Models** (HMMs): Key turning point in the mid-1980s
  - Shift from template matching based approach to a statistical approach
  - First described in 60’s, first used in 70’s, gained popularity in the 80’s
- Frederick Jelinek at IBM’s Watson Research Center pioneered the data driven approach to speech processing
  - "*Every time I fire a linguist, the performance of the speech recognizer goes up.*"
- With the popularity of statistical models, the field started agreeing upon standardized benchmarks and shared datasets

90’s – consumer ASR systems

Technology | CYBERTIMES The New York Times ON THE WEB

Home | Site Index | Site Search | Forums | Archives | Marketplace

April 3, 1997

New Software Greatly Advances Computer Dictation

By STEPHEN C. MILLER

Continuous speech has been the Holy Grail of the voice/speech recognition industry for years, and while it hasn't quite been achieved yet, Dragon Systems Inc. unveiled a new computer dictation system on Wednesday that represents a significant step forward in that quest.

"Everybody knows that we will be able to talk to our computers," the actor Richard Dreyfuss told a news conference in New York announcing the new system, known as Naturally Speaking. "But the question has always been: when?" Dreyfuss, who identified himself as an unpaid, unofficial spokesman for the Newton, Mass. company ("I'm just a fan"), implied that the when was now.

Whither speech recognition: the next 25 years

Publisher: IEEE Cite This PDF

D.B. Roe ; J.G. Wilpon **All Authors**

Published in: IEEE Communications Magazine (Volume: 31 , Issue: 11, November 1993)

Milestones in speech recognition and understanding [1]

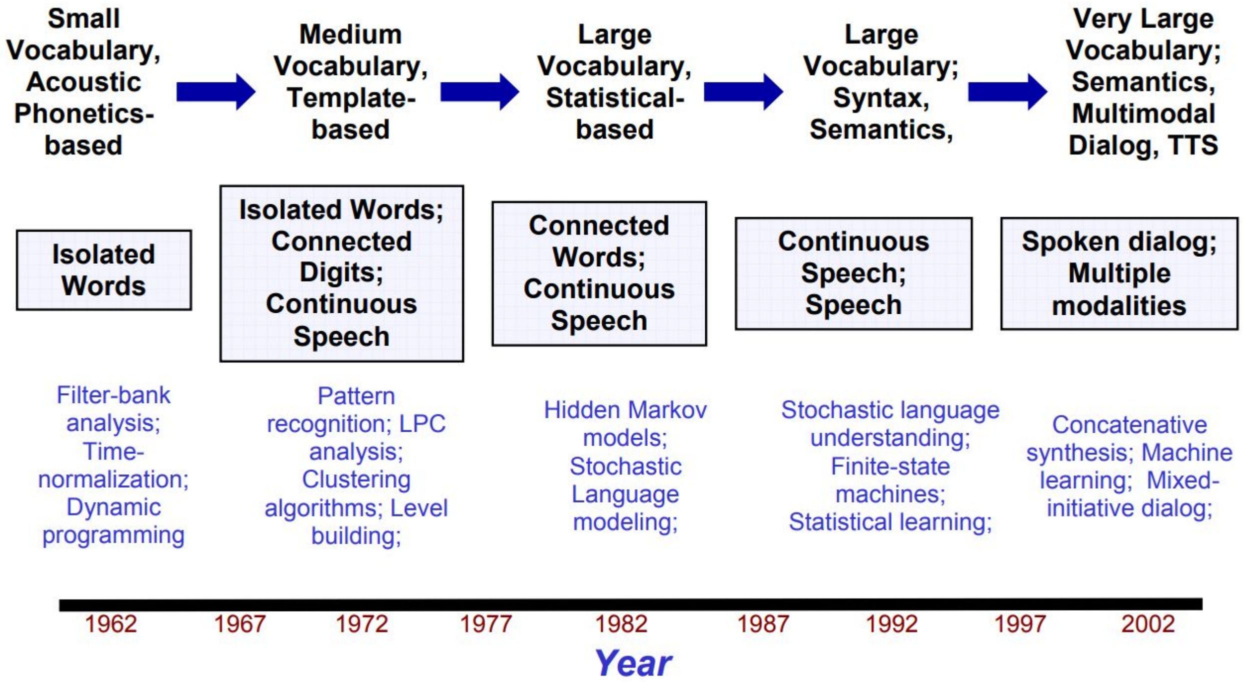


Figure 10 Milestones in Speech Recognition and Understanding Technology over the Past 40 Years.

[1] Juang, B. H., & Rabiner, L. R. (2005). Automatic speech recognition–a brief history of the technology development. Georgia Institute of Technology. Atlanta Rutgers University and the University of California. Santa Barbara, 1(67), 1.

How to represent the spoken content?

Breaking down words

- In ASR, a model is not trained to perform a ~50,000 class (words) classification
- Basically, an input sequence is taken and an output sequence is produced
- Each word is represented as a phoneme, a set of elementary sounds in a language based on the **International Phonetic Alphabet (IPA)**.
- In 1888, the IPA was invented in order to have a system in which there was a one-to-one correspondence between each sound in language and each phonetic symbol
    - E.g. French as / f ɹ ɛ n t ʃ /
- There are around 40 to 50 different phonemes in English.

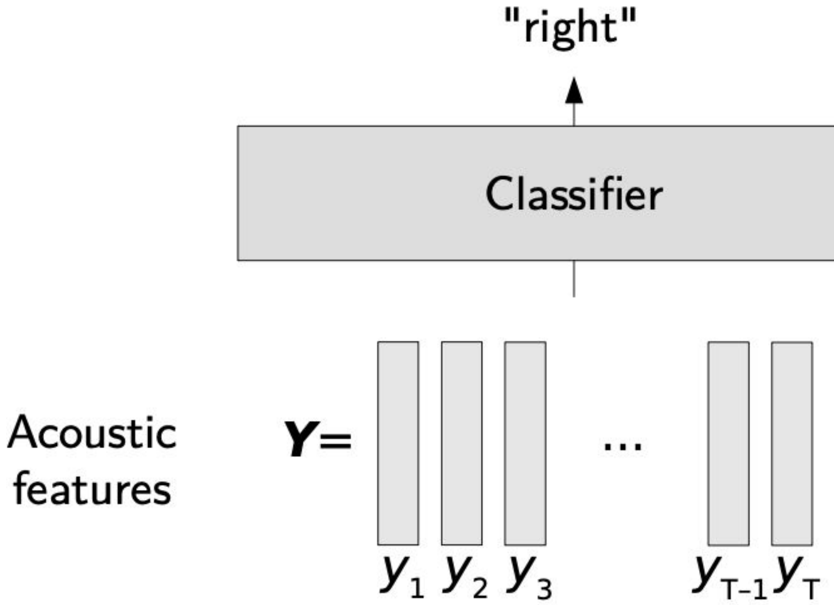
How to represent a word?

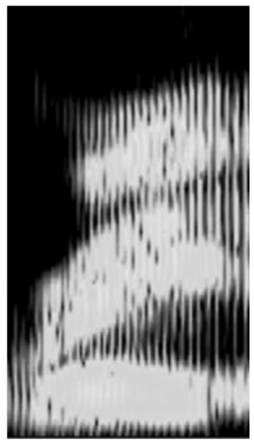
- As a sequence of morphemes (meaningful units of a language)
  - E.g contradiction can be broken up as contra-dict-ion
- As a part-of-speech (POS) in morpho-syntax: grammatical class
  - E.g noun, verb, etc. and flexional information, e.g singular, plural, gender…
- As a syntax describing the function of the word (subject, object, etc.)
- As a meaning
- As a sequence of phonemes
- As a sequence of graphemes (mostly a written symbol representing phonemes)

Labeling of speech

- Labeling speech can be done at several levels:
  - Sequence of words
  - Sequence of phones
- And the labels may be time-aligned if we know when they occur in speech.
- **Vocabulary** – the set of unique words in a specific ASR task
  - Large vocabulary continuous speech recognition (LVCSR)
  - Out Of Vocabulary words (OOV) – words never been seen in training
- Two types of speech recognition tasks:
  - Isolated word recognition
  - Continuous speech recognition

Isolated word recognition

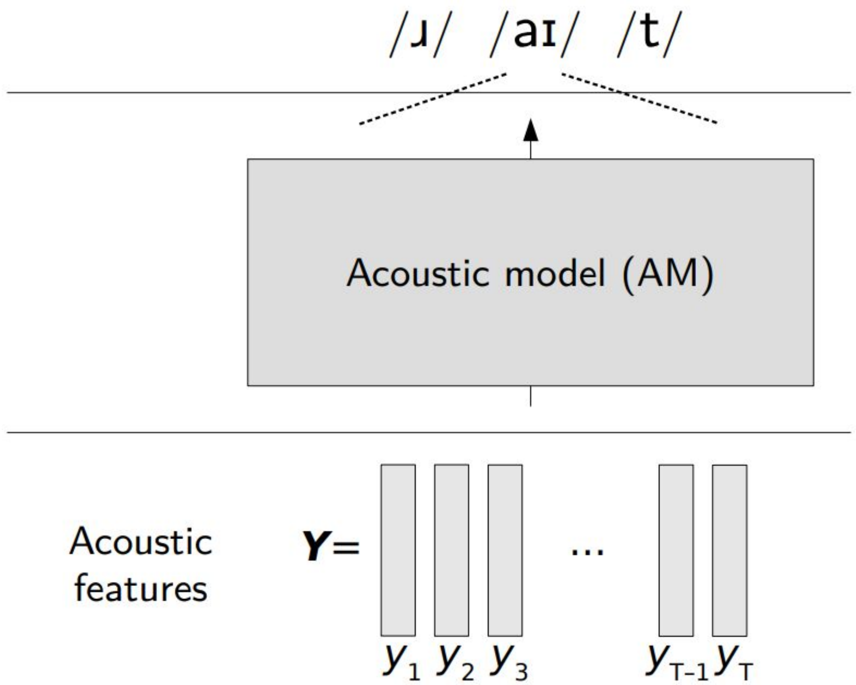


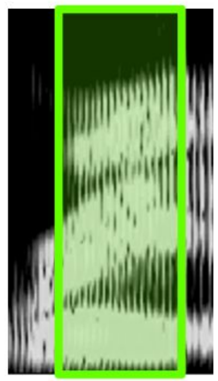


Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

Continuous speech recognition

- Decomposition into phonemes





Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

Performance evaluation

Evaluation metrics

- We usually evaluate the performance of an ASR system using **Word Error Rate (WER)**.
- We take as a reference a manual transcript.
- We then compute the number of mistakes made by the ASR system.
- Mistakes might include:
  - **Substitutions**, N_{SUB}, a word gets replaced
  - **Insertions**, N_{INS}, a word which was not pronounced in added
  - **Deletions**, N_{DEL}, a word is omitted from the transcript

Evaluation metric

- The WER is computed as:

$$WER = \frac{N_{SUB} + N_{INS} + N_{DEL}}{|N_{words-transcript}|}$$

- The perfect WER should be as close to 0 as possible.
- The number of substitutions, insertions and deletions is computed using the **Wagner-Fischer** dynamic programming algorithm for word alignment
  - Cost-effective solution compared to the recursive **Levenshtein Distance**

Levenshtein distance (LD)

- LD between “kitten” and “sitting” is 3
- At a minimum, 3 edits are required to change one into the other.
  - kitten → sitten (substitution of “s” for “k”)
  - sitten → sittin (substitution of “i” for “e”)
  - sittin → sitting (insertion of “g” at the end).

Levenshtein distance (LD)

- LD between “kitten” and “sitting” is 3
- At a minimum, 3 edits are required to change one into the other.
  - kitten → sitten (substitution of “s” for “k”)
  - sitten → sittin (substitution of “i” for “e”)
  - sittin → sitting (insertion of “g” at the end).

$$\operatorname{lev}_{a,b}(i,j) = \begin{cases} \max(i,j) & \text{if } \min(i,j)=0, \\ \min \begin{cases} \operatorname{lev}_{a,b}(i-1,j)+1 \\ \operatorname{lev}_{a,b}(i,j-1)+1 \\ \operatorname{lev}_{a,b}(i-1,j-1)+1_{(a_i \neq b_j)} \end{cases} & \text{otherwise.} \end{cases}$$

Word Error Rate

Evaluation: Word Error Rate (WER)

- **Reference:** manual transcript `the lazy dog jumps`
- **Hypothesis:** ASR output `amazing dog jumps`
- Alignment

```
the     lazy dog jumps
*** amazing dog jumps
```

- Editings

```
Del     Sub
```

- Score: $\mathrm{WER} = \frac{N_{\mathrm{Ins}} + N_{\mathrm{Del}} + N_{\mathrm{Sub}}}{|\mathrm{Reference}|}$ Edit distance
  - Perfect $= 0\%$
  - can be $> 100\%$ (many insertions) `(0+1+1)/4 = 50%`

Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

Phoneme Error Rate (PER)

- **REF**: **recognise speech** ⟵⟶ ɹ e k ə **g** n aɪ̯ **z** s pi: t ʃ
- **HYP**: **wreck a nice beach** ⟵⟶ ɹ e k **eɪ** n aɪ̯ s **bi:** t ʃ
- **Substitutions** = ?
- **Insertions** = ?
- **Deletions** = ?
- **PER** = ?

Phoneme Error Rate (PER)

- **REF**: **recognise speech** ⟷ ɹ e k ə **g** n aɪ̯ **z** s pi: t ʃ
- **HYP**: **wreck a nice beach** ⟷ ɹ e k **eɪ** n aɪ̯ s **bi:** t ʃ
- **Substitutions** = 2
- **Insertions** = 0
- **Deletions** = 2
- **PER** = (2+0+2)⁄12 = 0.33 ~ 33%

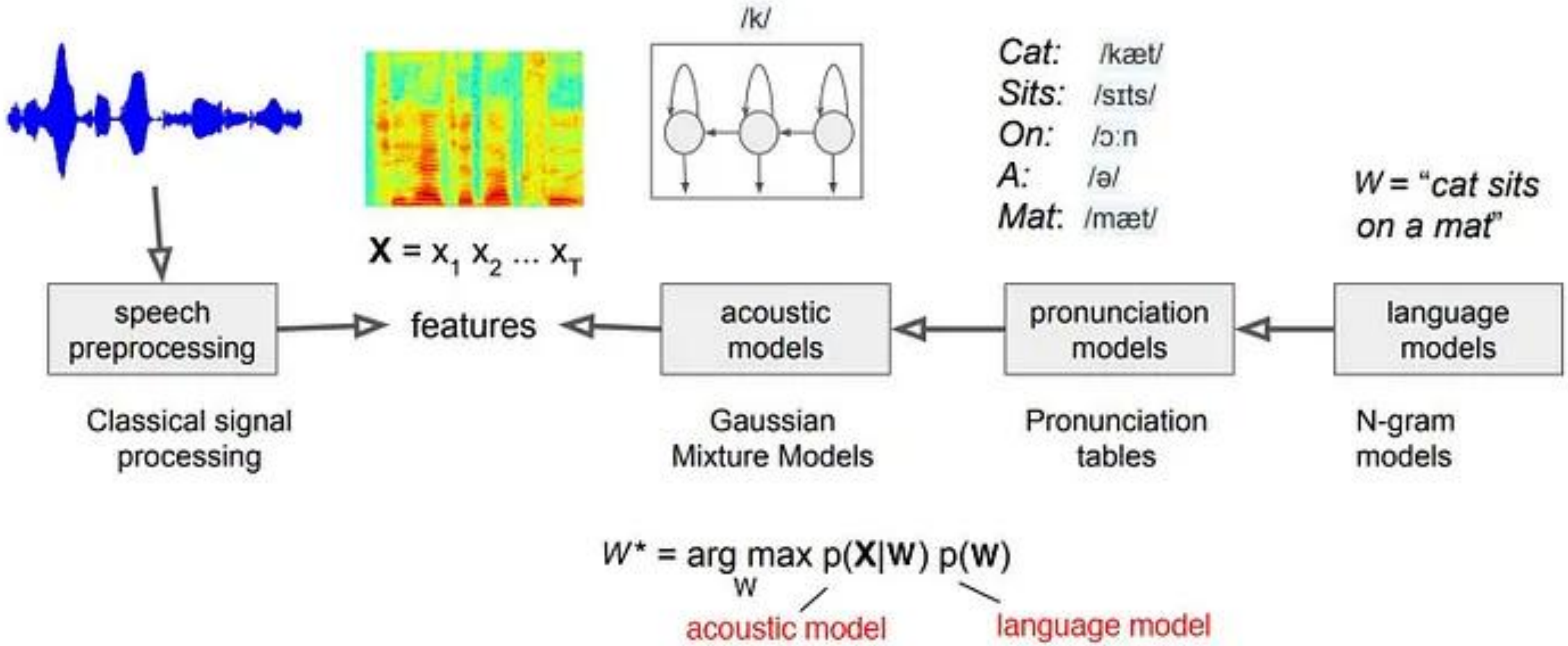
Example results

Table 1. Reported results on TIMIT core test set.

Method	Frame error rate	Phoneme error rate
HMM [20]	39.3	42.0
HMM [11]	35.1	40.9
KSBSC [11]	-	45.1
PA [16]	30.0	33.4
DROP [16]	29.2	31.1
PAC-Bayes 1-frame	**27.7**	**30.2**
Online LM-HMM [20]	25.0	30.2
Batch LM-HMM [4]	-	28.2
CRFs (9-frames MLP) [21]	-	29.3
PAC-Bayes 9-frames	**26.5**	**28.6**

J. Keshet, D. McAllester and T. Hazan, "PAC-Bayesian approach for minimization of phoneme error rate," 2011 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), Prague, Czech Republic, 2011, pp. 2224-2227

Speech Recognition - the classical way



Picture courtesy: https://jonathan-hui.medium.com/speech-recognition-gmm-hmm-8bb5eff8b196

Statistical Speech Recognition

Fundamental Equation of Statistical Speech Recognition

- Given, a sequence of acoustic features X, and an optimal word sequence ***W**** from the vocabulary (*among all possible **W***), we aim to find:

$$W^\star = argmax_W P(W \mid X)$$

- Using Bayes rule:

$$W^\star = argmax_W \frac{P(X \mid W)P(W)}{P(X)}$$

- Removing the constant denominator:

$$W^\star = argmax_W P(X \mid W)P(W)$$

Fundamental Equation of Statistical Speech Recognition

- Given, a sequence of acoustic features X, and an optimal word sequence ***W**** from the vocabulary (*among all possible **W***), we aim to find:

$$W^{\star} = argmax_W P(W \mid X)$$

- Using Bayes rule:

$$W^{\star} = argmax_W \frac{P(X \mid W)P(W)}{P(X)}$$

Acoustic Model

- Removing the constant denominator:

$$W^{\star} = argmax_W P(X \mid W)P(W)$$

Fundamental Equation of Statistical Speech Recognition

- Given, a sequence of acoustic features X, and an optimal word sequence ***W**** from the vocabulary (*among all possible **W***), we aim to find:

$$W^{\star} = argmax_W P(W \mid X)$$

- Using Bayes rule:

$$W^{\star} = argmax_W \frac{P(X \mid W)P(W)}{P(X)}$$

Acoustic Model

- Removing the constant denominator:

Language Model

$$W^{\star} = argmax_W P(X \mid W)P(W)$$

Fundamental Equation of Statistical Speech Recognition

- Given, a sequence of acoustic features X, and an optimal word sequence ***W**** from the vocabulary (*among all possible **W***), we aim to find:

$$W^\star = argmax_W P(W \mid X)$$

- Using Bayes rule:

$$W^\star = argmax_W \frac{P(X \mid W)P(W)}{P(X)}$$

Acoustic Model

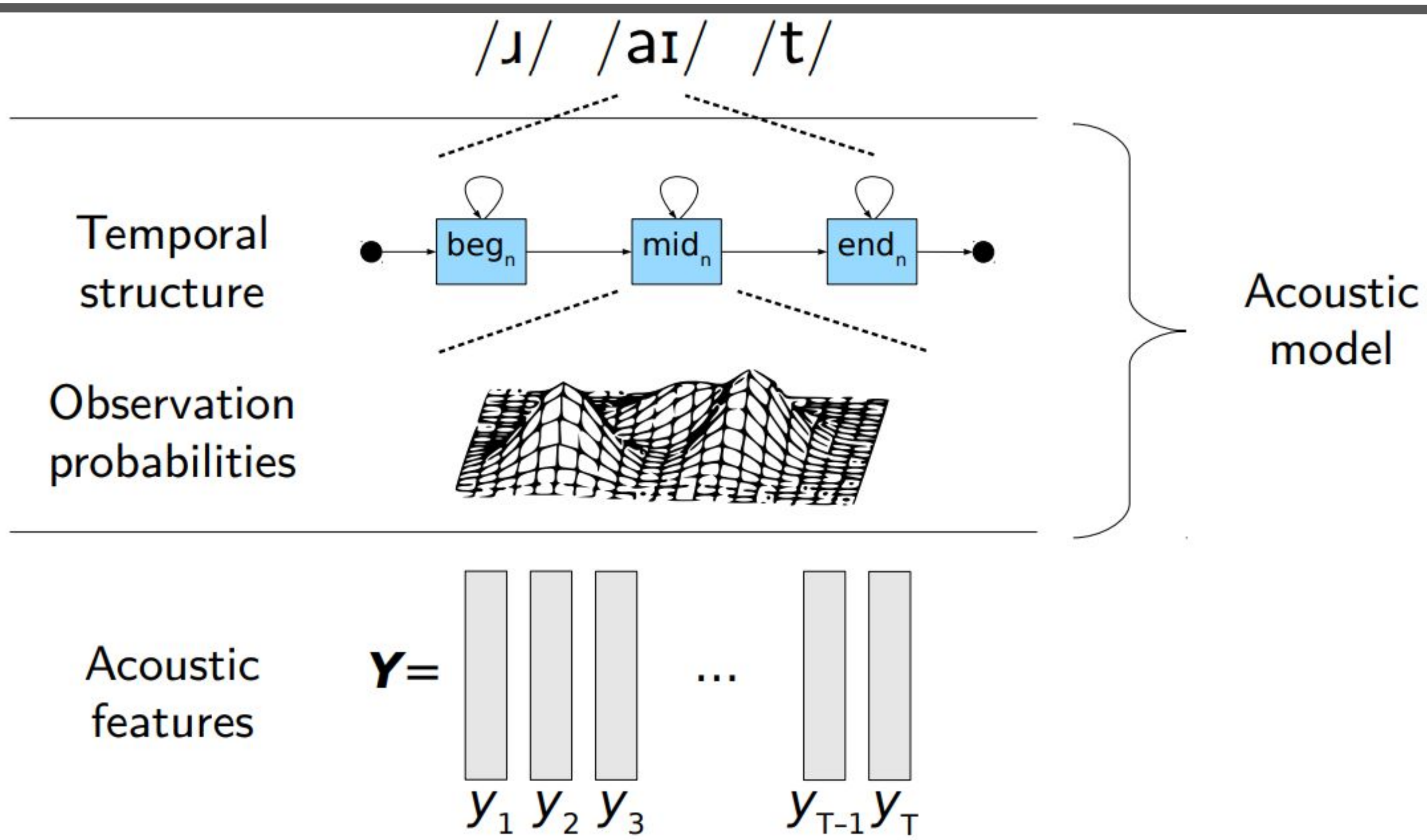
Search Space

Language Model

- Removing the constant denominator:

$$W^\star = argmax_W P(X \mid W)P(W)$$

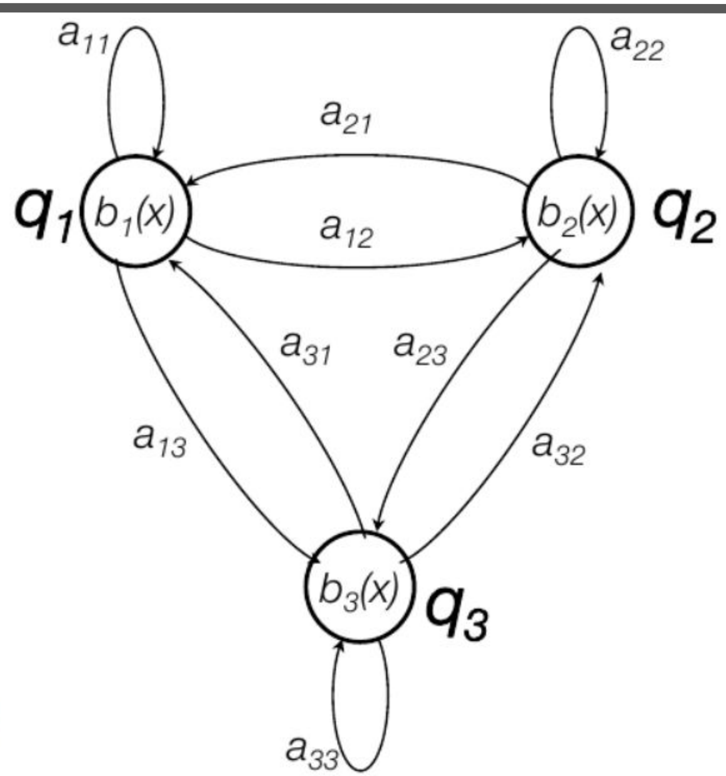
Hidden Markov Models (HMMs)



Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

HMMs

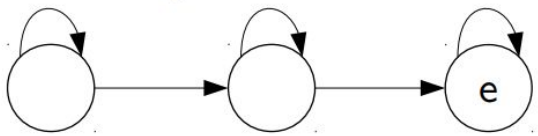
- Probabilistic automaton
  - States q_i
    - State at step t: s_t
    - Inititial probability $\pi_i = \Pr(s_0 = q_i)$
    - Transition probabilities $a_{ij} = \Pr(s_{t+1} = q_j \mid s_t = q_i)$
  - Outputs
    - Observation at step t: o_t
    - Alphabet of symbols $\mathcal{X} = (x_k)$
    - Emission probability $b_i(x) = \Pr(o_t = x \mid s_t = q_i)$



Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

HMMs

- 1 phoneme
  - 3 (or 5-)-state linear HMM : beginning, middle, end
  - Observation independence
    - $\Pr(o_t = x \mid s_t = q_i, s_{t-1} = q_{i'}, \ldots, s_0 = q_{i''})$
      $= \Pr(o_t = x \mid s_t = q_i)$
  - Probability to reach the end state e at frame y_t



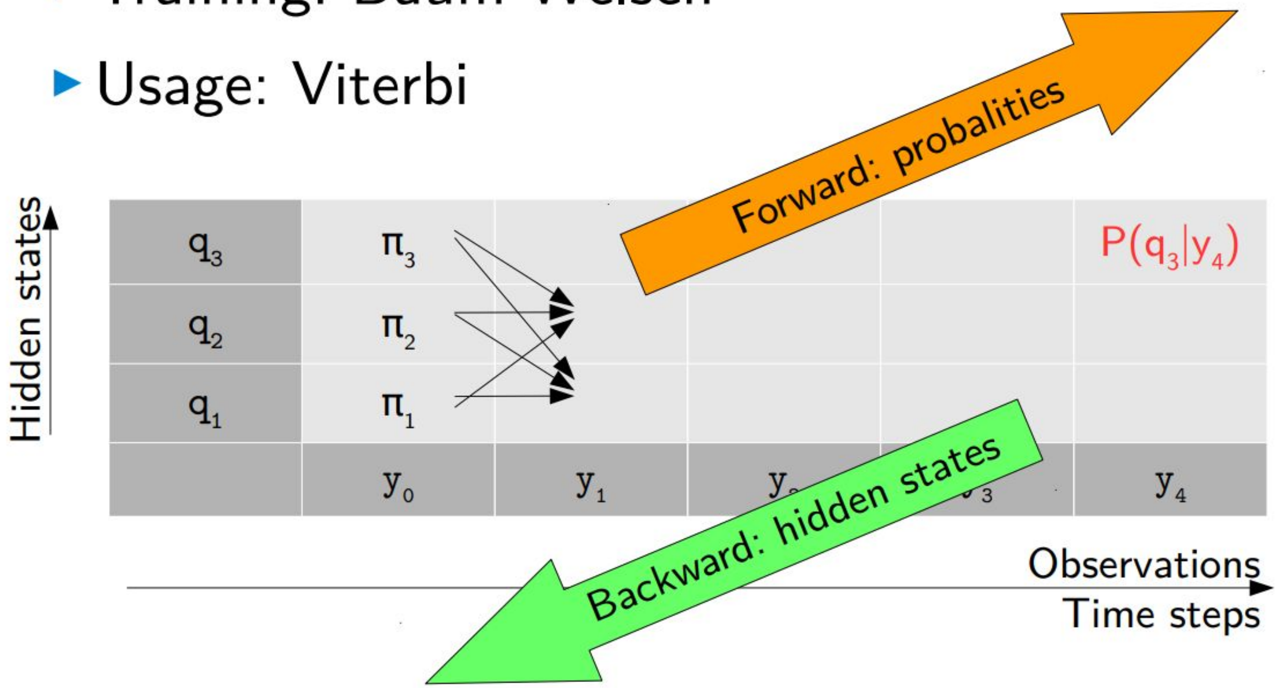
$$P(/a/|t) = P(s_t = e)P(b_e = y_t)$$

- Context-dependent phoneme = triphones
  - Same linear HMMs
  - E.g., /bab/, /bal/, /bap/, etc.
  - State-tying: gather similar HMM state to overcome data sparsity

Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

HMMs

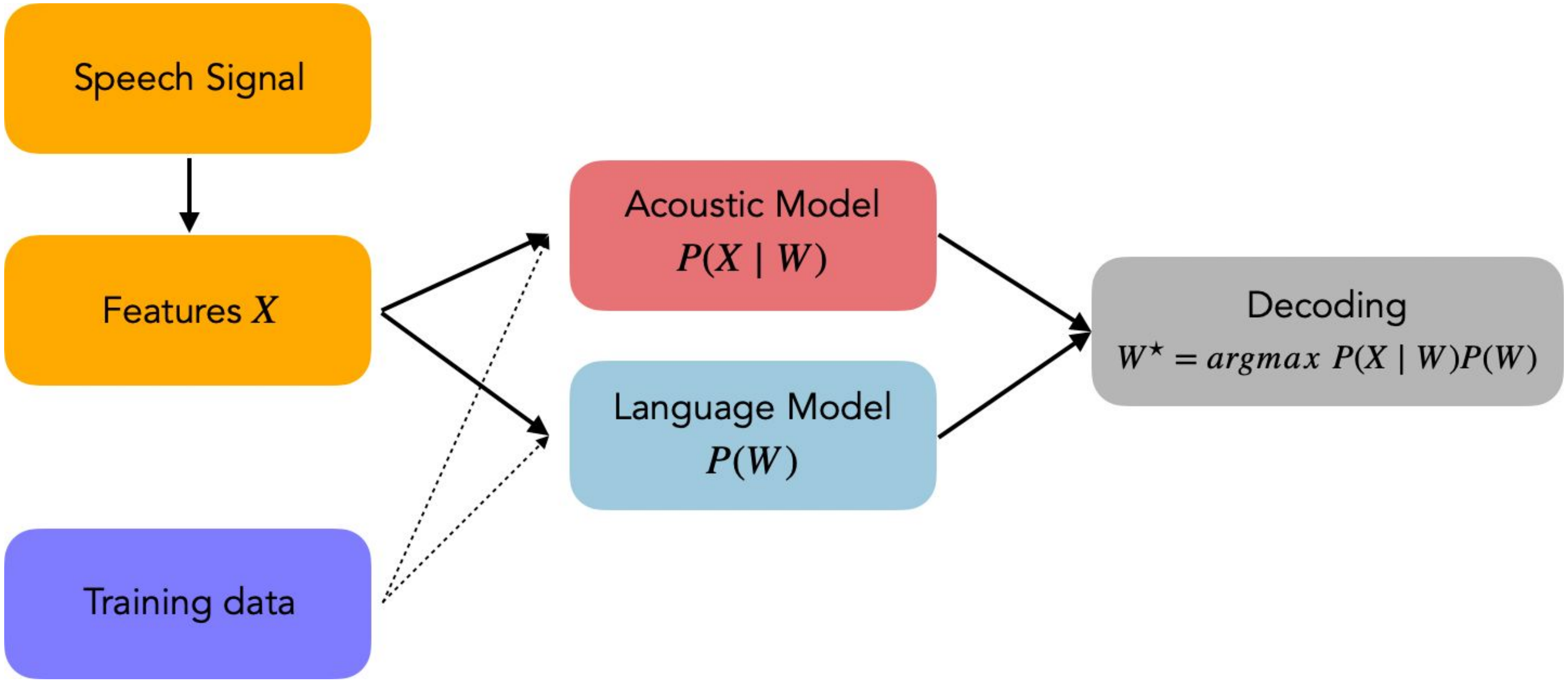
- Training: Baum-Welsch
- Usage: Viterbi



Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

Classical ASR systems

Cascaded ASR pipeline



Picture courtesy: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr

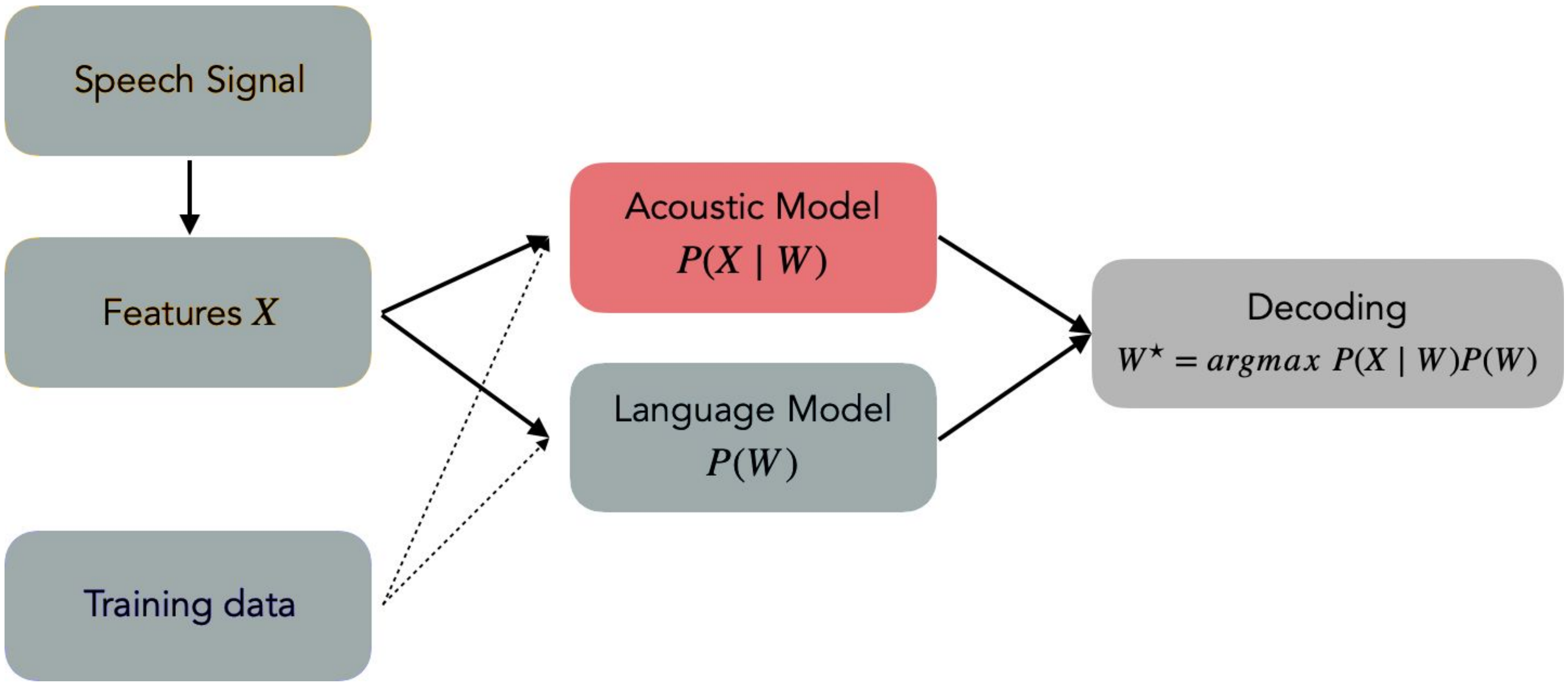
Classical ASR systems

Acoustic Models

Feature extraction ***X***

- We should extract features ***X*** which are:
  - robust across speakers
  - robust against noise and channel effects
  - low dimension, at equal accuracy
  - non-redundant among features
- Features we typically extract include:
  - Mel-Frequency Cepstral Coefficients (MFCC), Perceptual Linear Prediction (PLP)
- Extracted features are generally normalized using **z-score normalization**
  - $X^* = (X - \mu)/\sigma$; $\mu \rightarrow$ mean, $\sigma \rightarrow$ standard deviation

Acoustic model



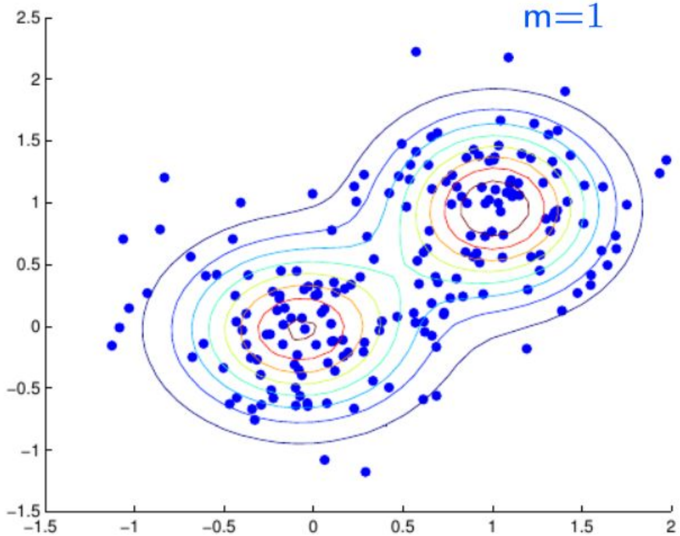
Adapted from: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr

Acoustic model

- It models the **relationship between the audio signal and the phonetic units** in the language.
- In isolated word/pattern recognition, the acoustic features (X) are used as an input to a classifier whose role is to output the correct word.
- In continuous speech recognition, we take input sequence and should output sequences too.

HMM-GMM acoustic model

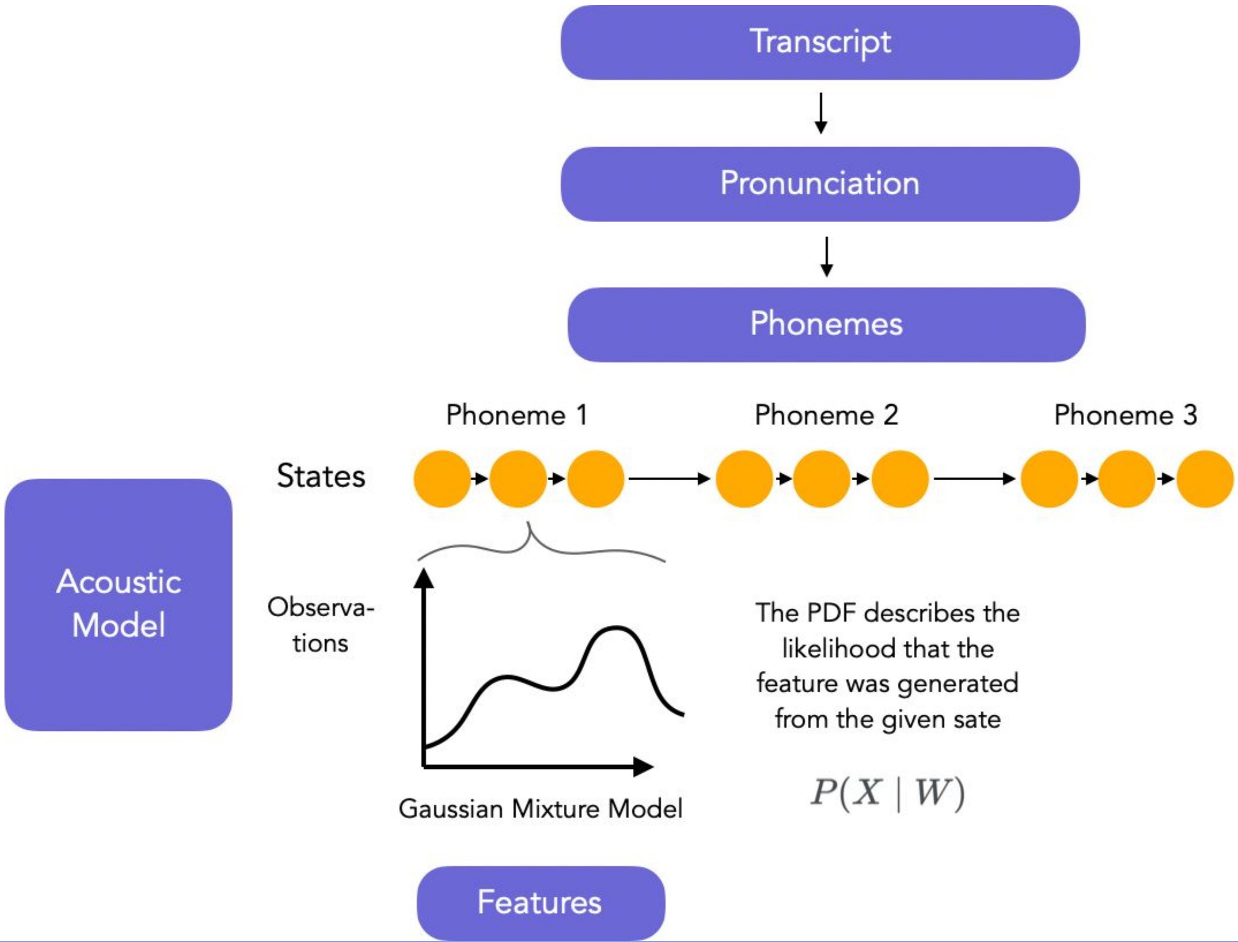
- Gaussian Mixture Model (GMM)
  - M components
  - $\Pr(o_t = \mathbf{y} \mid s_t = q_i) = b_i(\mathbf{y}) = \sum_{m=1}^{M} \mathcal{N}(\mathbf{y}, \boldsymbol{\mu}_{jm}, \boldsymbol{\Sigma}_{jm})$



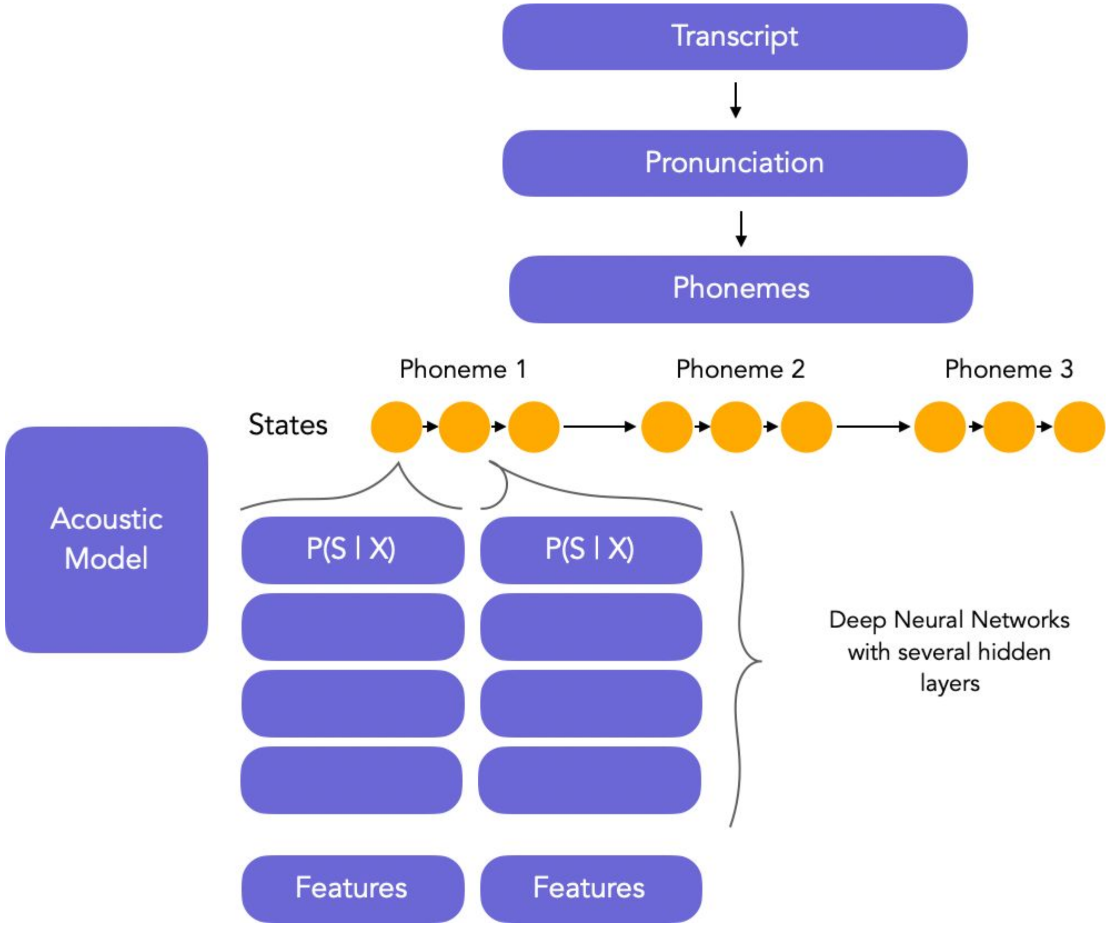
Picture courtesy: https://people.irisa.fr/Gwenole.Lecorve/lectures/ASR.pdf

HMM-GMM acoustic model

Picture courtesy: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr



HMM-DNN acoustic models



Picture courtesy: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr

HMM-DNN acoustic models

- Latest models focus on hybrid HMM-DNN architectures and approach the acoustic model in another way.
- In such approach, we do not care about the acoustic model $P(X|W)$, but we directly tackle $P(W|X)$, as the probability of observing state sequences given X.
- The aim of the DNN is to model the posterior probabilities over HMM states.

HMM-DNN acoustic models

- DNN produces posterior, whereas the Viterbi Backward-Forward algorithm requires P(X|W) (from the GMM-HMM acoustic model).
- Therefore, we use Bayes Rule:

$$P(X \mid W) = \frac{P(W \mid X)P(X)}{P(W)}$$

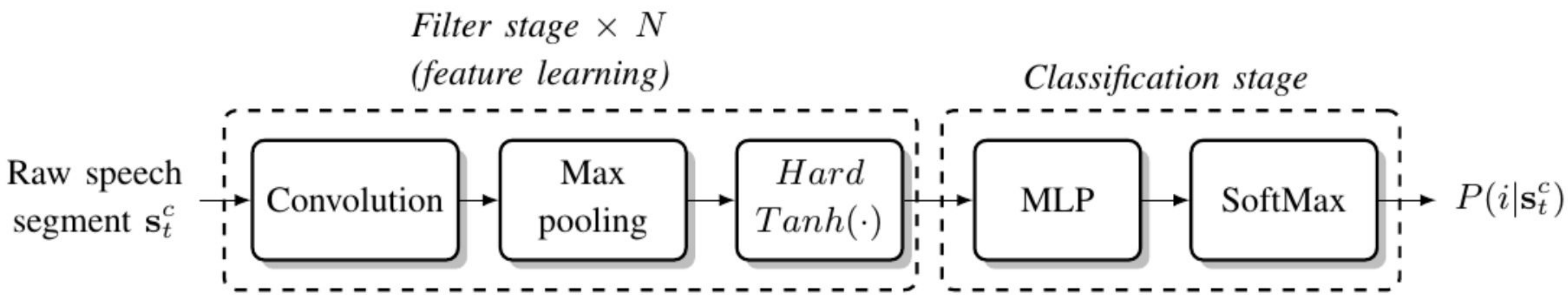
- The probability of the acoustic feature P(X) is not known, but it just scales all the likelihoods by the same factor, and therefore does not modify the alignment.

HMM/DNN vs. HMM/GMM

HMM/DNN	HMM/GMM
Considers short term correlation	Assumes no correlation in inputs
No probability distribution function assumption	Assumes GMMs as PDFs
Discriminative training in the generated distributions	No discriminative training in the generated distributions (can be overlapping)
Discriminative acoustic model at frame level	Poor discrimination (Maximum Likelihood instead of Maximum A Posteriori)
Higher performance	Lower performance

End-to-end acoustic models

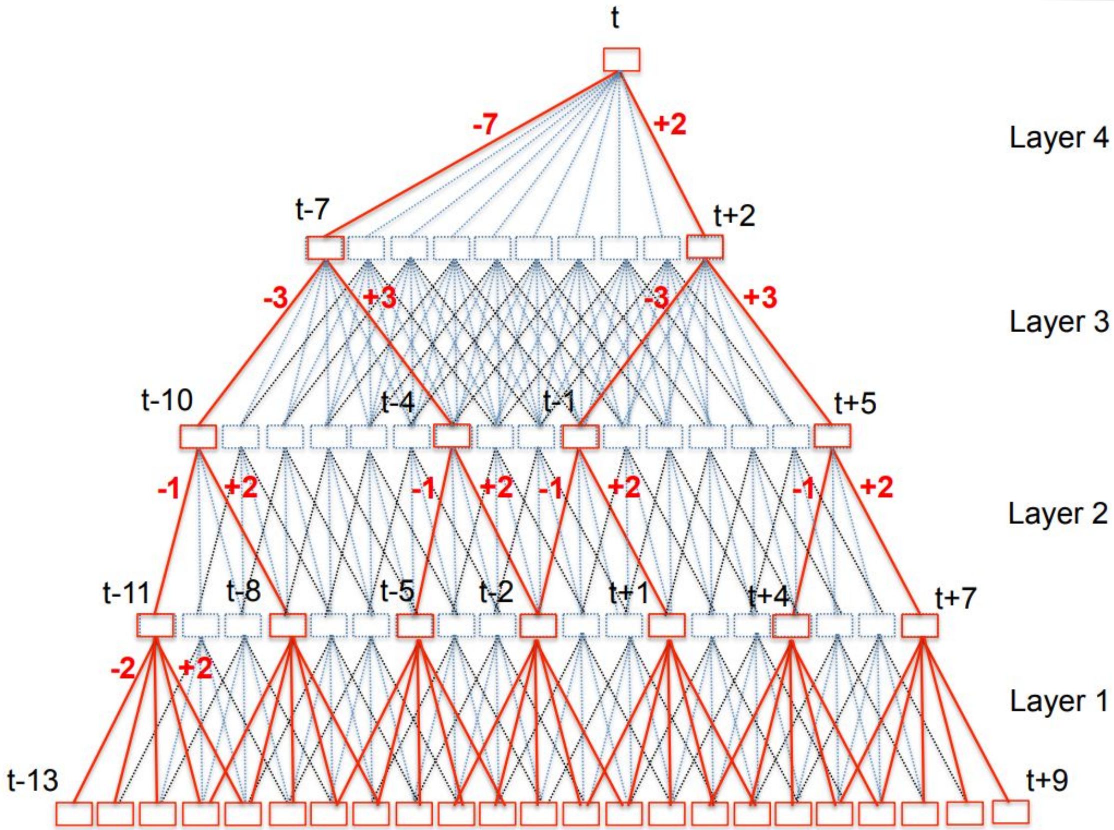
- In End-to-end models, the steps of feature extraction and phoneme prediction are combined:



Picture courtesy: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr

Time Delay Neural Networks (TDNN)

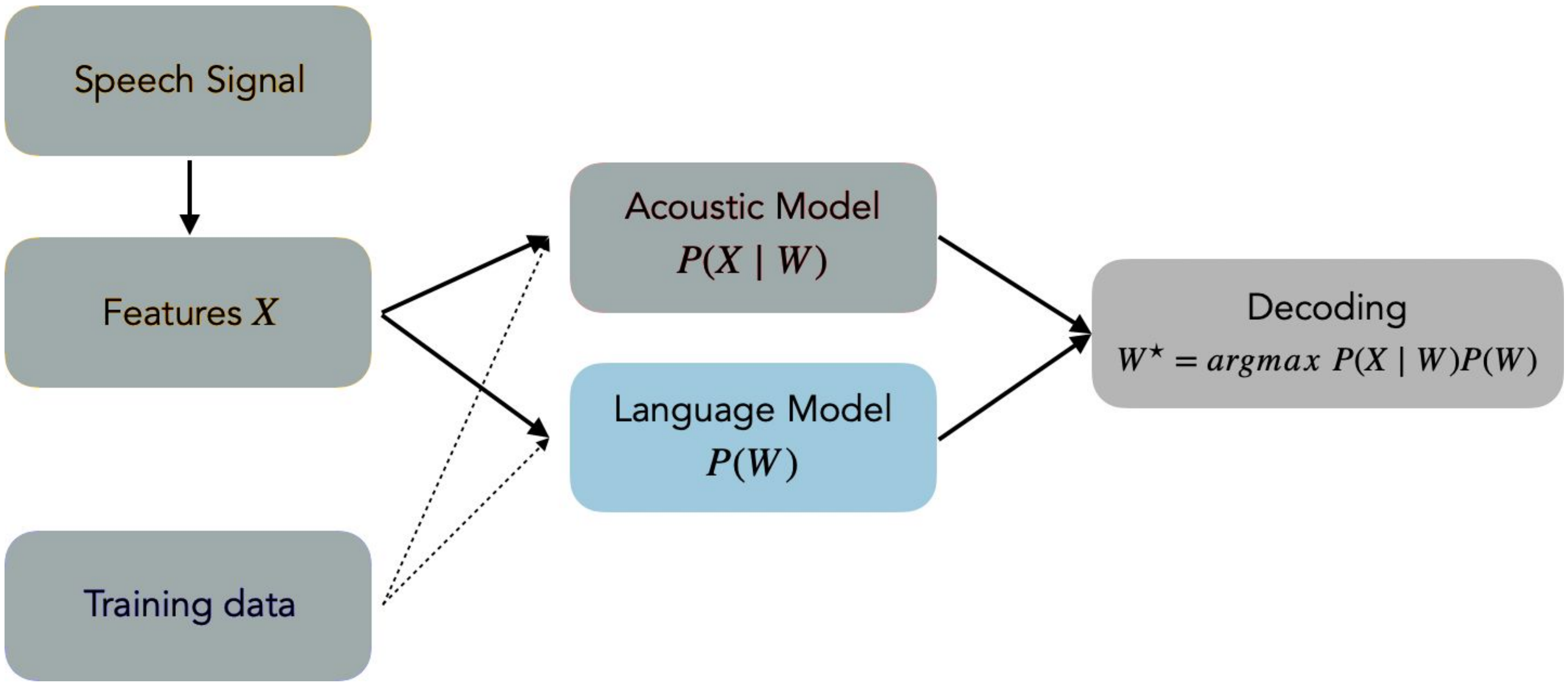
Picture courtesy: Peddinti, V., Povey, D., Khudanpur, S. (2015) A time delay neural network architecture for efficient modeling of long temporal contexts. Proc. Interspeech 2015, 3214-3218



Classical ASR systems

Language Models

Language model



Adapted from: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr

Pronunciation model

$$W^\star = argmax_W P(W \mid X)$$

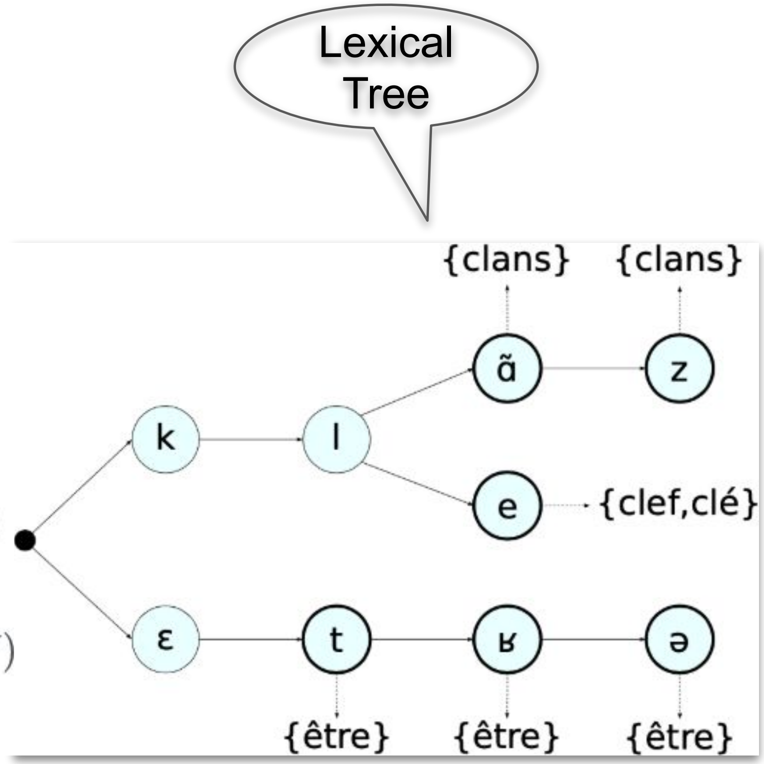
We consider words are being sequences of states Q.

$$W^\star = argmax_W P(X \mid Q, W) P(Q, W)$$

$$W^\star \approx argmax_W P(X \mid Q) \sum_Q P(Q \mid W) P(W)$$

$$W^\star \approx argmax_W max_Q P(X \mid Q) P(Q \mid W) P(W)$$

Where $P(Q \mid W)$ is the **pronunciation model**.



Picture courtesy: https://maelfabien.github.io/machinelearning/speech_reco/#what-is-asr

- The pronunciation dictionary is written by human experts, and defined in the IPA
- Grapheme-to-phoneme (G2P) models try to learn automatically the pronunciation of new words

Language Modelling

- The language model is defined as P(W)
- The constraint on P(W) is that ∑WP(W)=1
- Statistical language modeling aims to disambiguate sequences such as: “recognize speech”, “wreck a nice beach”
- The maximum likelihood estimation of a sequence is given by:

$$P(w_i \mid w_1, \ldots, w_{i-1}) = \frac{C(w_1, \ldots, w_i)}{\sum_v C(w_1, \ldots, w_{i-1} v)}$$

Language Modelling

Observed Count

- The language model is defined as P(W)
- The constraint on P(W) is that ∑WP(W)=1
- Statistical language modeling aims to disambiguate sequences such as: “recognize speech”, “wreck a nice beach”
- The maximum likelihood estimation of a sequence is given by:

$$P(w_i \mid w_1, \ldots, w_{i-1}) = \frac{C(w_1, \ldots, w_i)}{\sum_v C(w_1, \ldots, w_{i-1} v)}$$

Relative Frequency

Language Modelling

Observed Count

- The language model is defined as P(W)
- The constraint on P(W) is that ∑WP(W)=1
- Statistical language modeling aims to disambiguate sequences such as: “recognize speech”, “wreck a nice beach”
- The maximum likelihood estimation of a sequence is given by:

$$P(w_i \mid w_1, \ldots, w_{i-1}) = \frac{C(w_1, \ldots, w_i)}{\sum_v C(w_1, \ldots, w_{i-1} v)}$$

Relative Frequency

- E.g.:
  - P(the|its water is so transparent that) = C(its water is so transparent that the)/C(its water is so transparent that)

Language Modelling

- Chain rule:

$$P(w_1, \ldots, w_N) = \prod_{k=1}^{N} (w_k \mid w_{k-1})$$

- This approach seems logical
  - But for longer the sequence, the probability most likely goes to 0

- Solutions:
  - Smoothing
  - Backoff

Smoothing and Backoff

- What if never observed during training?
  - $\rightarrow$ The longer sequence, the more zero-counts
- History of words

$$P(w_1, w_2, \ldots w_N) \approx P(w_1) \times \prod_{i=2}^{N} P(w_i | \Phi(w_1 \ldots w_{i-1})$$

- Smoothing
  - Redistribute probability mass from observed to unobserved events : change counts and renormalize
    - Absolute discouting, Kneser-Ney smoothing
- Backoff
  - Link unseen events to the most related seen events

N-Gram language model

- Truncate the word history

$$P(w_i|h) = P(w_i|w_{\mathbf{i\text{-}n+1}} \ldots w_{i-1})$$

with n usually 2..5

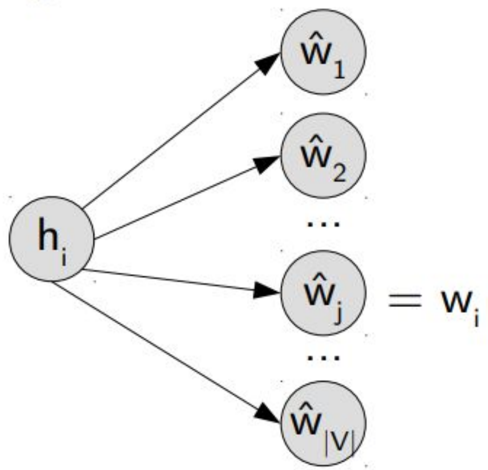
- $n = 1 \rightarrow$ "unigram", $n = 2 \rightarrow$ "bigram", $n = 3 \rightarrow$ "trigram"
- Backoff
  - Unseen $w_a w_b w_c \rightarrow$ fallback to $w_b w_c$
  - $P(w_c|\underbrace{w_a w_b}_{h}) = P(w_c|\underbrace{w_b}_{h^-})\ \beta(w_a w_b)$

Evaluating a Language Model – Perplexity

- How well a text T is predicted by a model M?
- Definition 1
  - Cross-entropy $H(P_T, P_M) = \sum_{w_i \in T} P_T(w_i) \log_2 P_M(w_i|h_i)$
  - Perplexity $PPL_M(T) = 2^{-H(P_T, P_M)}$
- Definition 2
  - Average log-likelihood of M over T (n words) $L(T|M) = \frac{1}{n} \times \sum_{w_i \in T} \log_2 P_M(w_i|h_i)$
  - Perplexity $PPL_M(T) = 2^{-L(T|M)}$

Perplexity

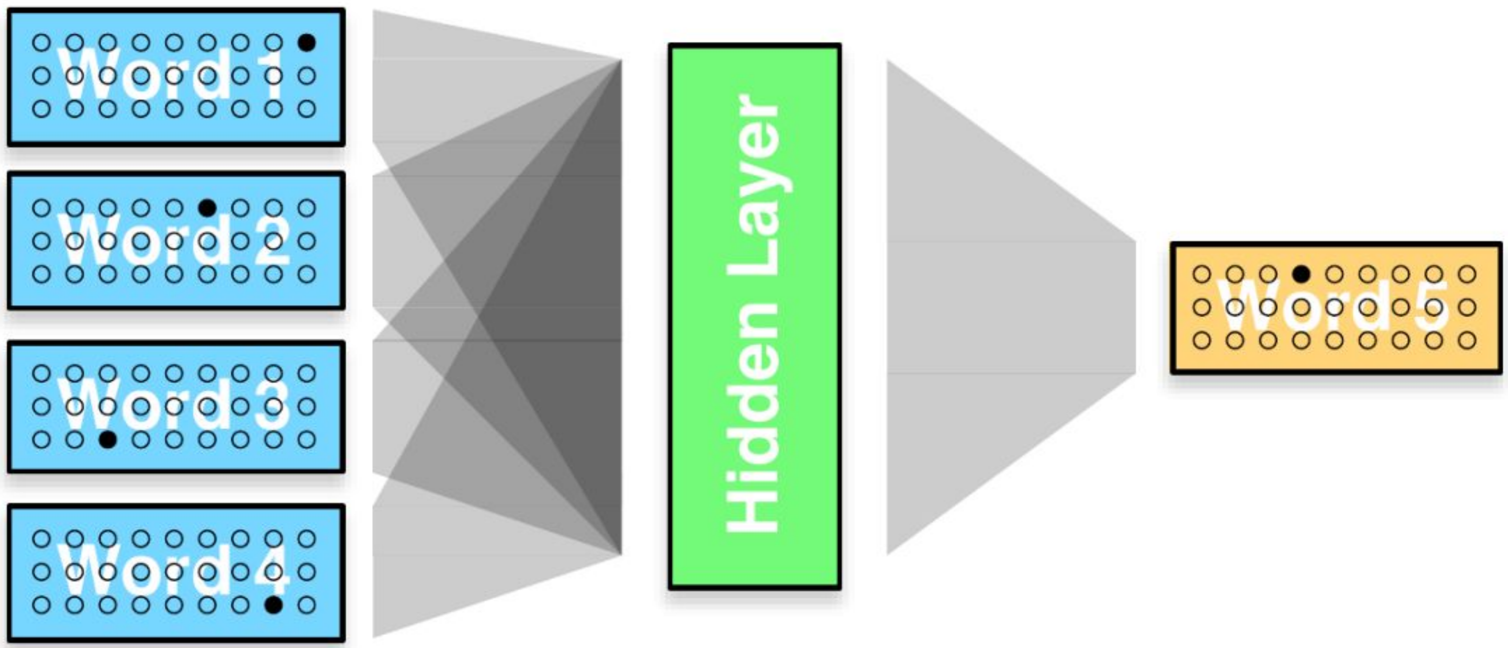
- The lower, the better
- Best theoretical perplexity $= 1$
- Intepretation
  - Branching factor



$E(j) = PPL$

Neural Network LMs

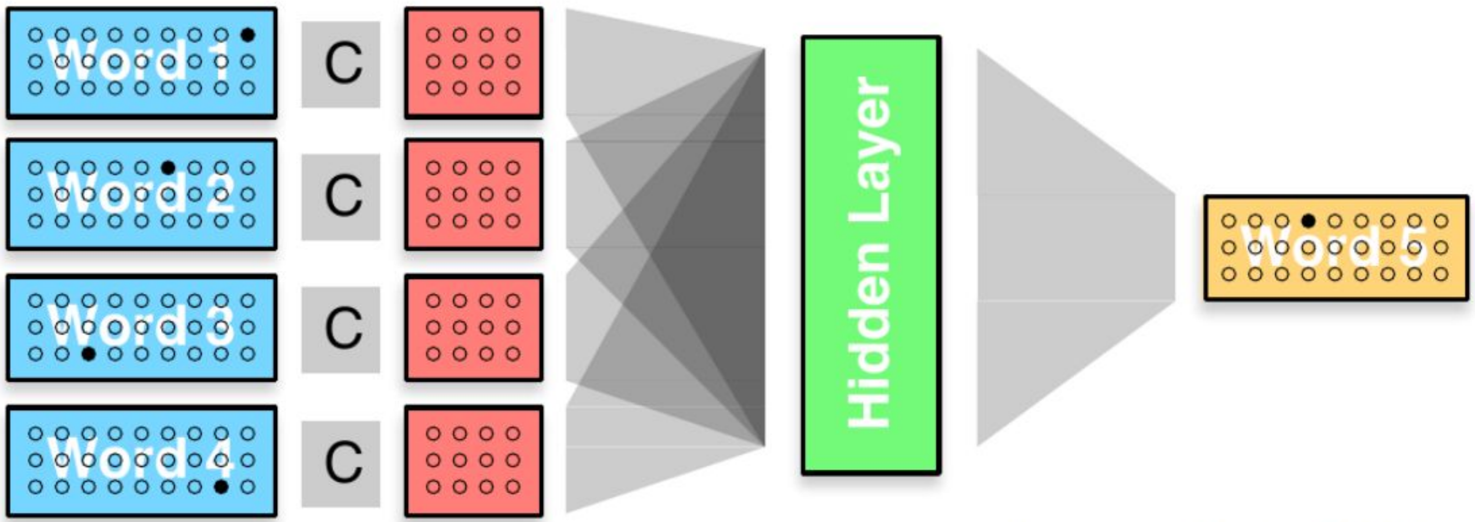
- Feed-forward approach



(source: Koehn, 2016)

Neural Network LMs – Word Embeddings

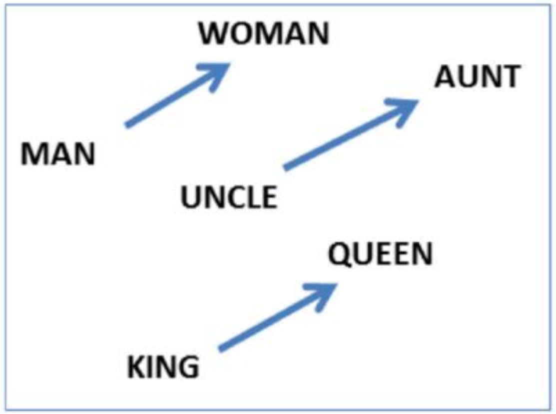
- Word embedding
- Shared weights C

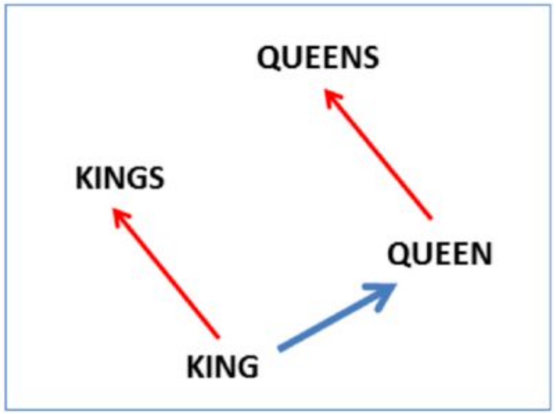


(source: Koehn, 2016)

Word embeddings

- Projection into a continuous space $\mathbb{R}^d$
- Topological properties
  - Semantics
  - Morpho-syntax





Performance comparison of various acoustic models

Model	Year	Test other
GMM/HMM	–	36.4
DNN/HMM	–	12.3
TDNN	2018	7.63

Classical ASR systems

Decoding

Decoding – Predicting the transcript or the Hypothesis

- Need: find

LM scale factor

$$W^* = \arg\max_{W} \underbrace{p(Y|W) \times P(W)^{\psi} \times I^{|W|}}_{\text{score}(W)}$$

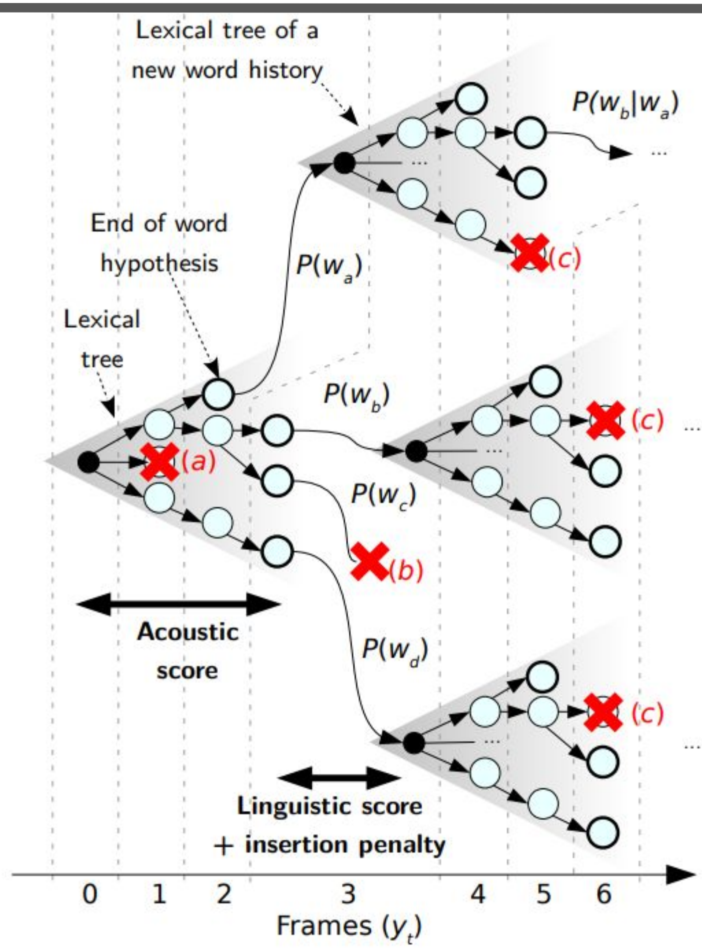
Word insertion penalty

while not exploring the whole search space

- Solution: beam search
  - Frame-synchronous (start at $t = 0$)
  - Idea: Parallel explorations limited a maximum of K active states
  - Advantage: memory and time efficient due to pruning of low interst partial hypotheses

Beam search decoding

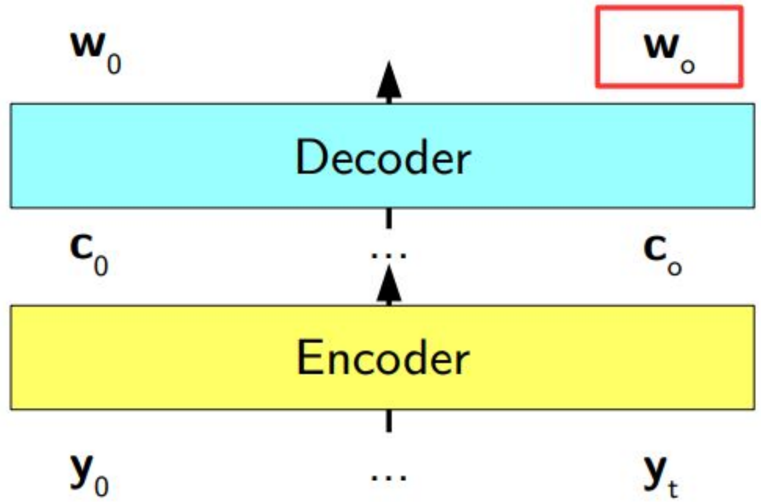
- Until end of word
  - Aggregate acoustics
- At end of word
  - Add LM score and insertion penalty
- At each step
  - Check the number of active states
    - → Pruning (a, b, c)



End-to-end approach for ASR decoding

$$\mathbf{W}^* = \arg\max_{\mathbf{W}} P(\mathbf{W}|\mathbf{Y})$$

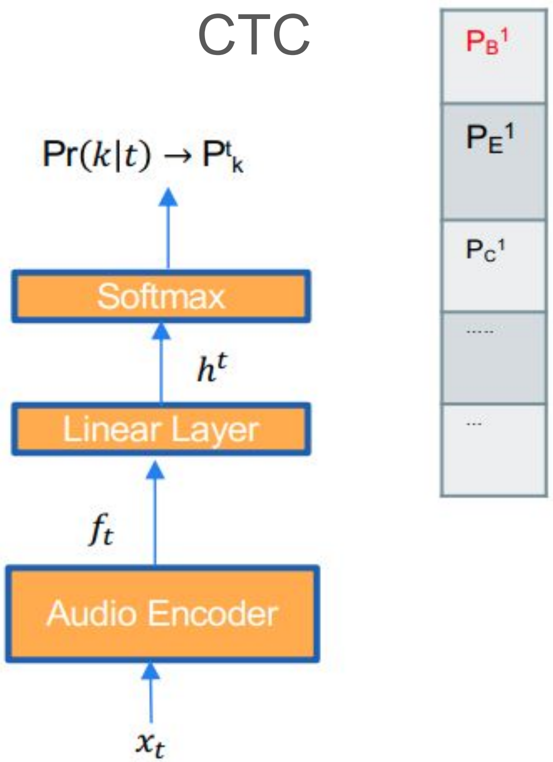
- Encode $\mathbf{Y}$ ($\mathbf{y}_t$) as a sequence of contexts $\mathbf{C}$ ($\mathbf{c}_t$)
- Decode $\mathbf{C}$ into a sequence of words $\mathbf{W}$

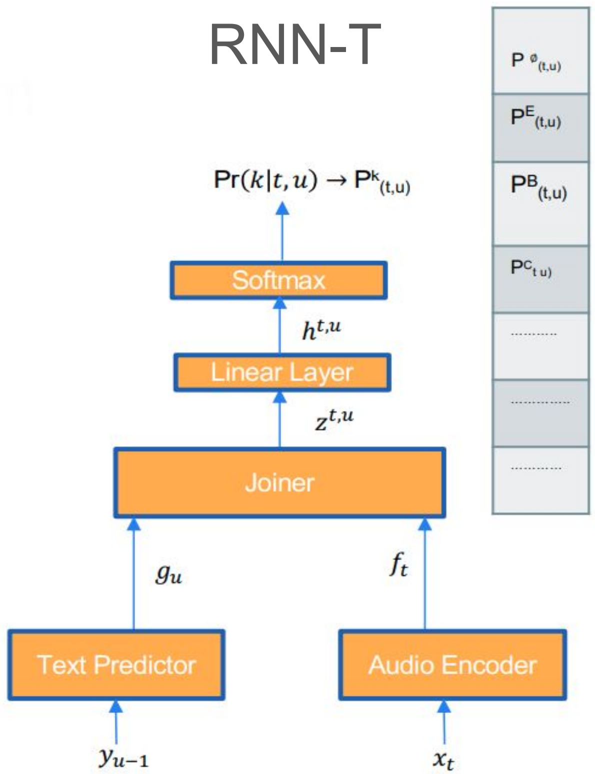


- Beam-search decoding

End-to-End ASR

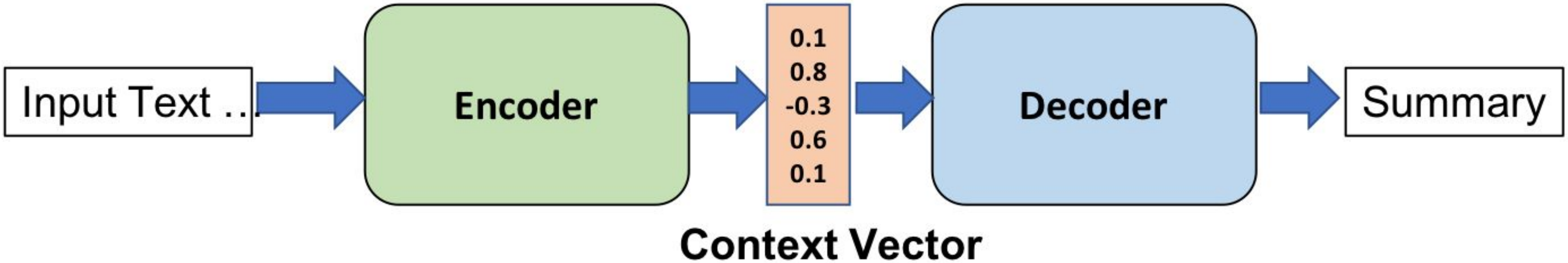
Popular E2E ASR losses [8]:





[8] L24, CS7643, AY2021, "**RNNT ASR Tutorial**", Georgia Tech, online: https://sites.cc.gatech.edu/classes/AY2021/cs7643_spring/assets/L24_rnnt_asr_tutorial_gt.pdf

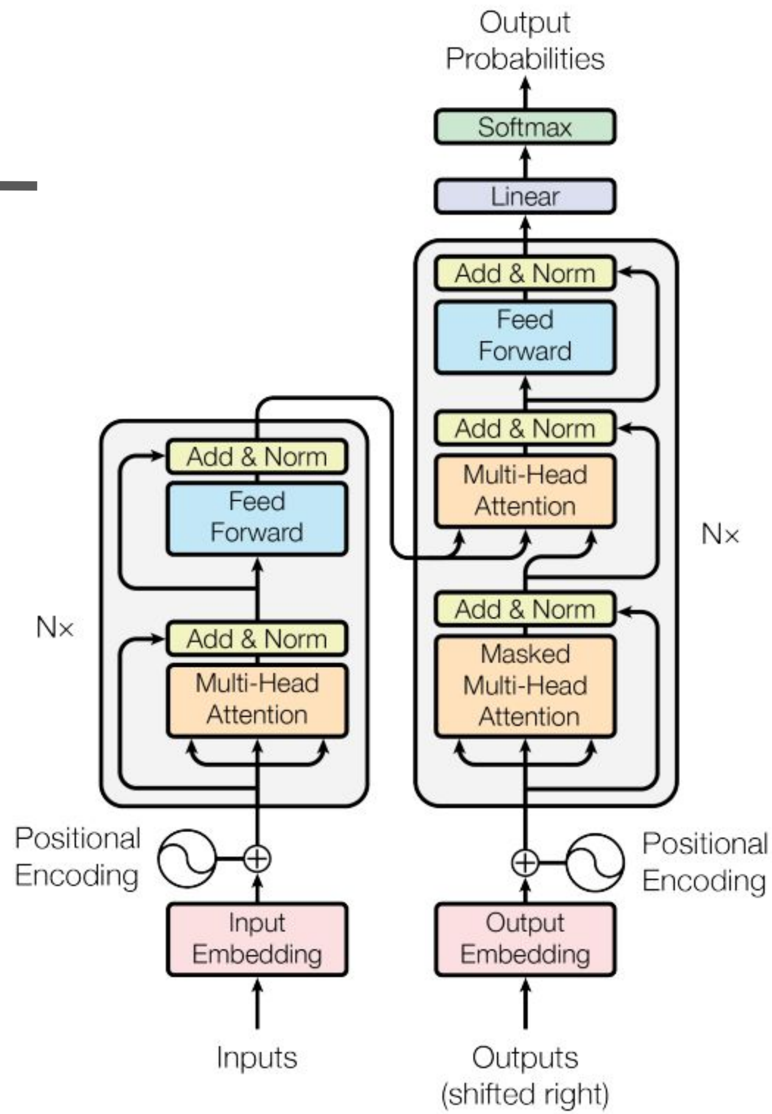
Encoder Decoder architecture



Picture courtesy: https://www.oreilly.com/library/view/hands-on-natural-language/9781789139495/7d9f1317-d2e0-46ea-b8f0-28d7b42ecabf.xhtml

Transformer architecture

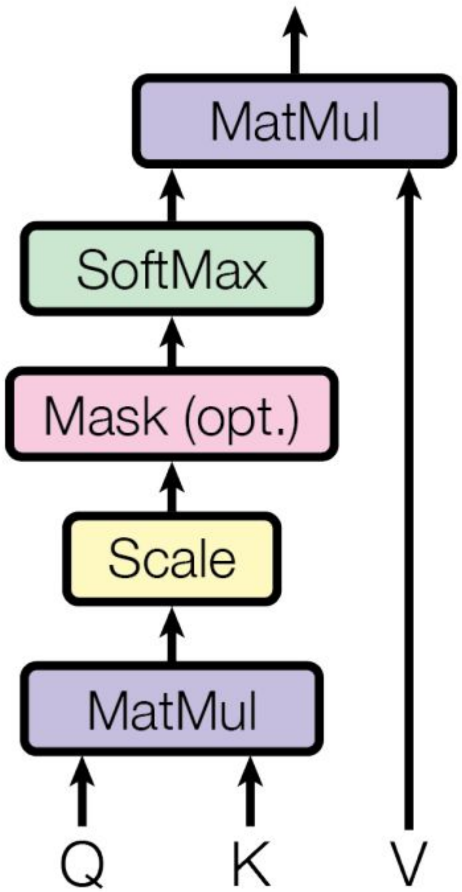
Picture courtesy: Vaswani, A. (2017). Attention is all you need. Advances in Neural Information Processing Systems.



Attention

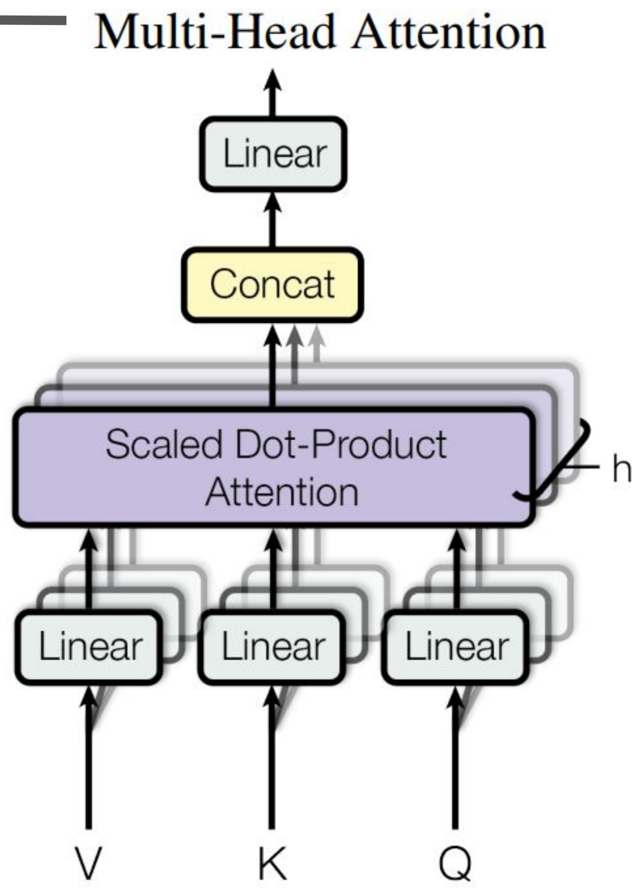
Picture courtesy: Vaswani, A. (2017). Attention is all you need. Advances in Neural Information Processing Systems.

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



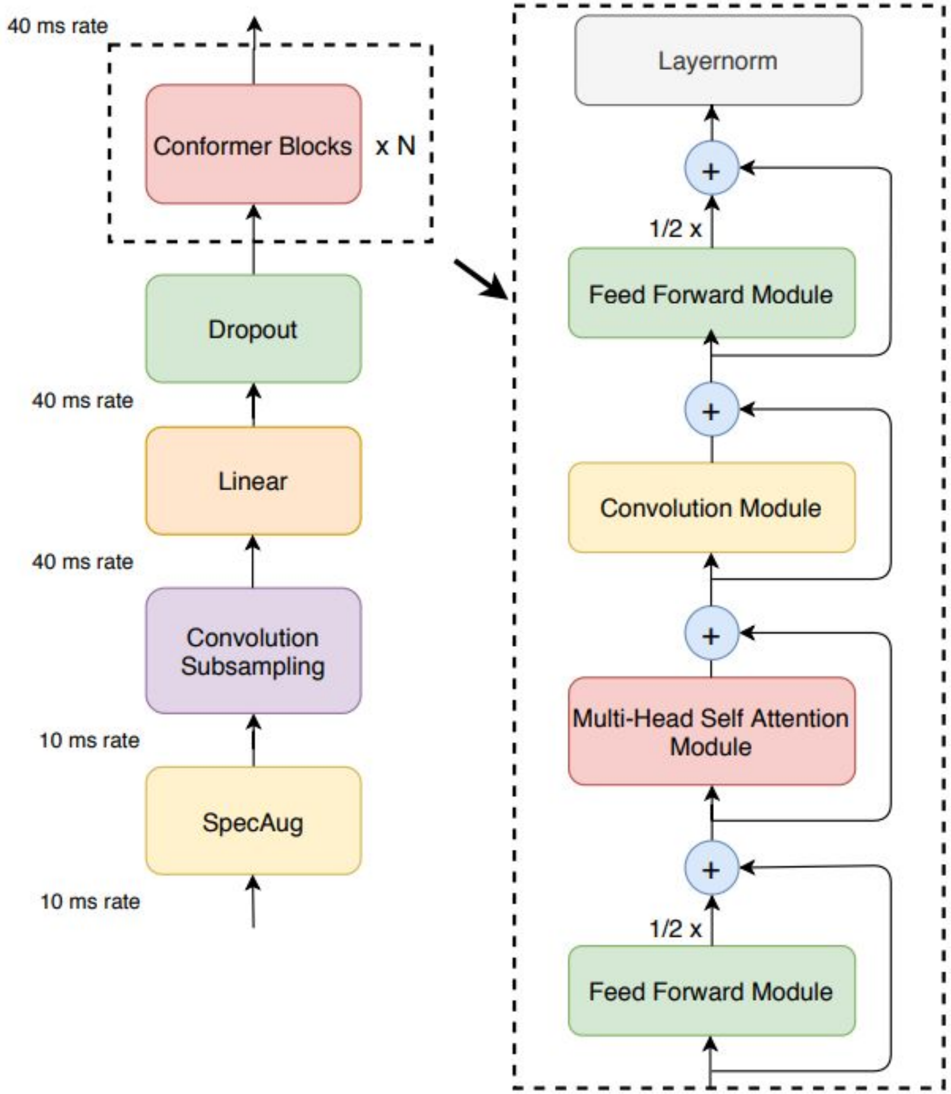
Multi Head Attention

Picture courtesy: Vaswani, A. (2017). Attention is all you need. Advances in Neural Information Processing Systems.

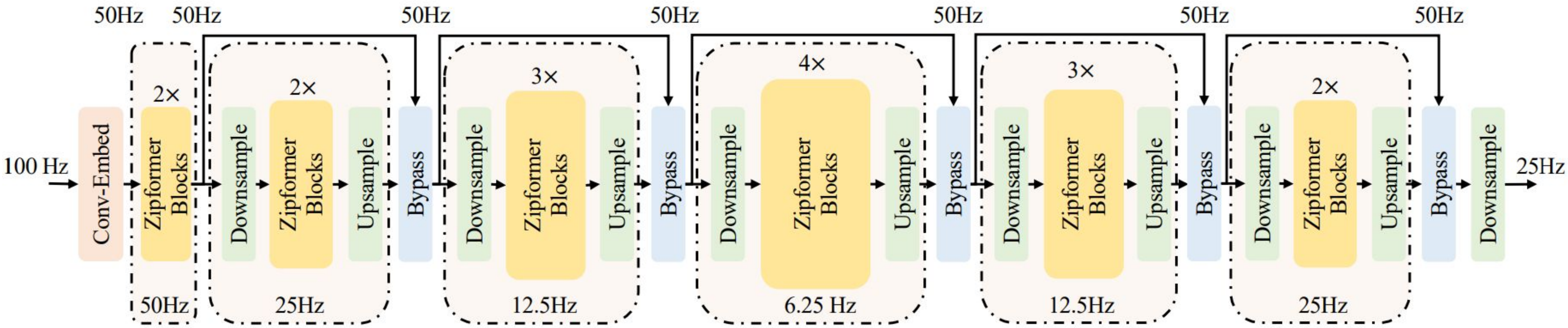


Conformer

Picture courtesy: Gulati, A., Qin, J., Chiu, C. C., Parmar, N., Zhang, Y., Yu, J., ... & Pang, R. (2020). Conformer: Convolution-augmented transformer for speech recognition. arXiv preprint arXiv:2005.08100.

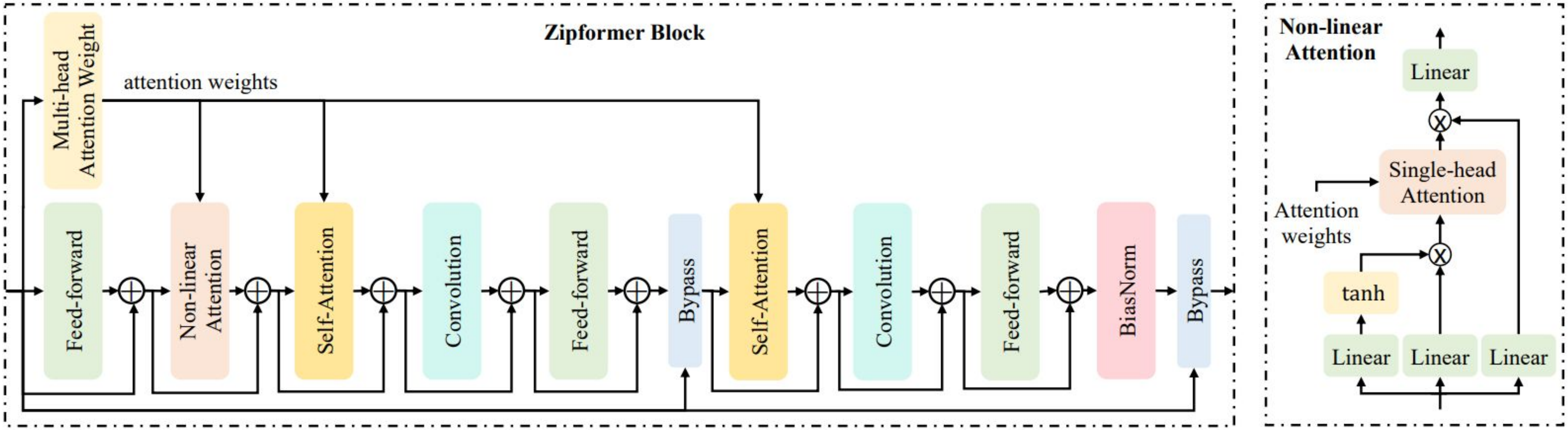


Zipformer



Picture courtesy: Yao, Z., Guo, L., Yang, X., Kang, W., Kuang, F., Yang, Y., ... & Povey, D. (2023, October). Zipformer: A faster and better encoder for automatic speech recognition. In The Twelfth International Conference on Learning Representations.

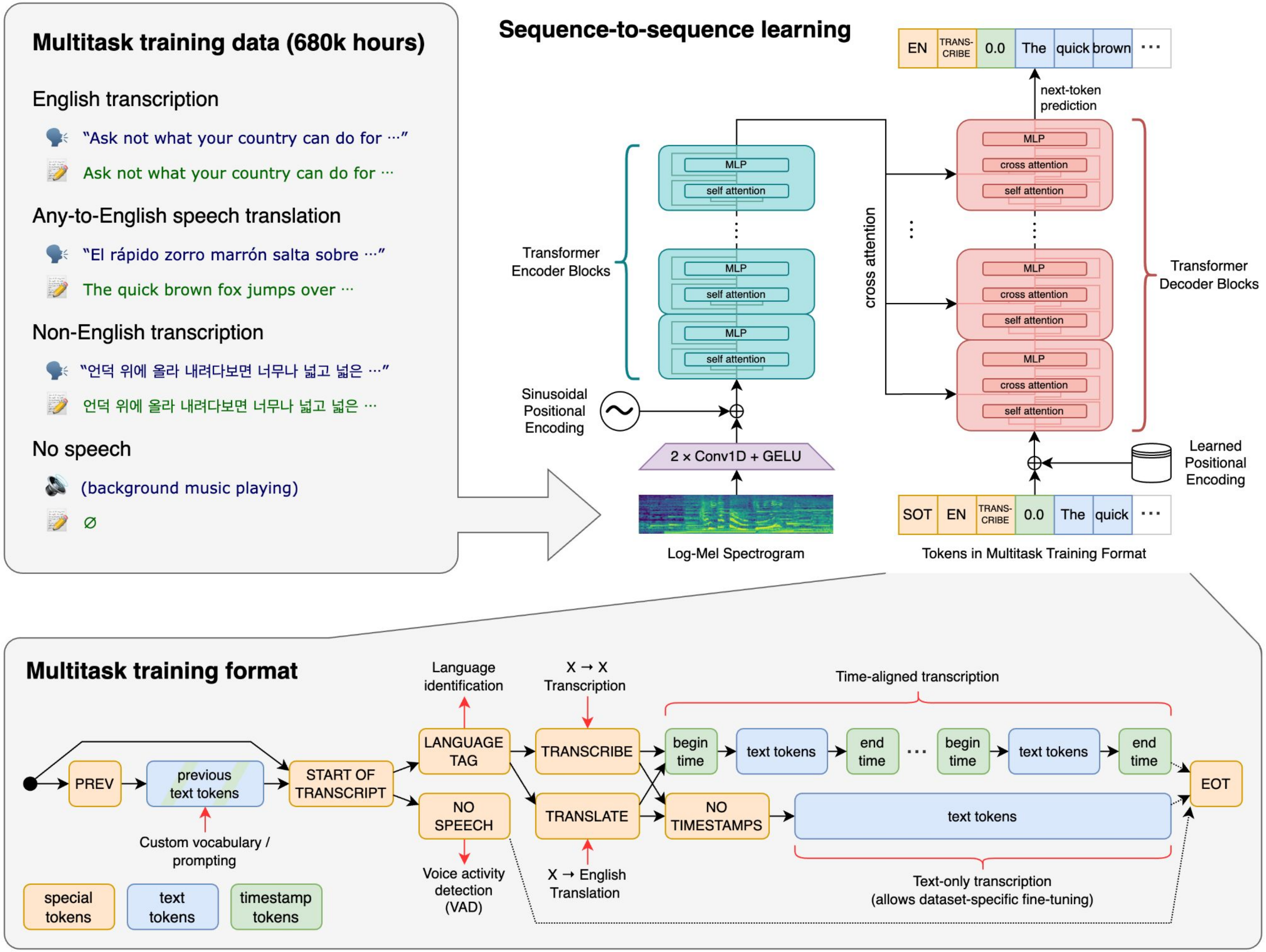
Zipformer



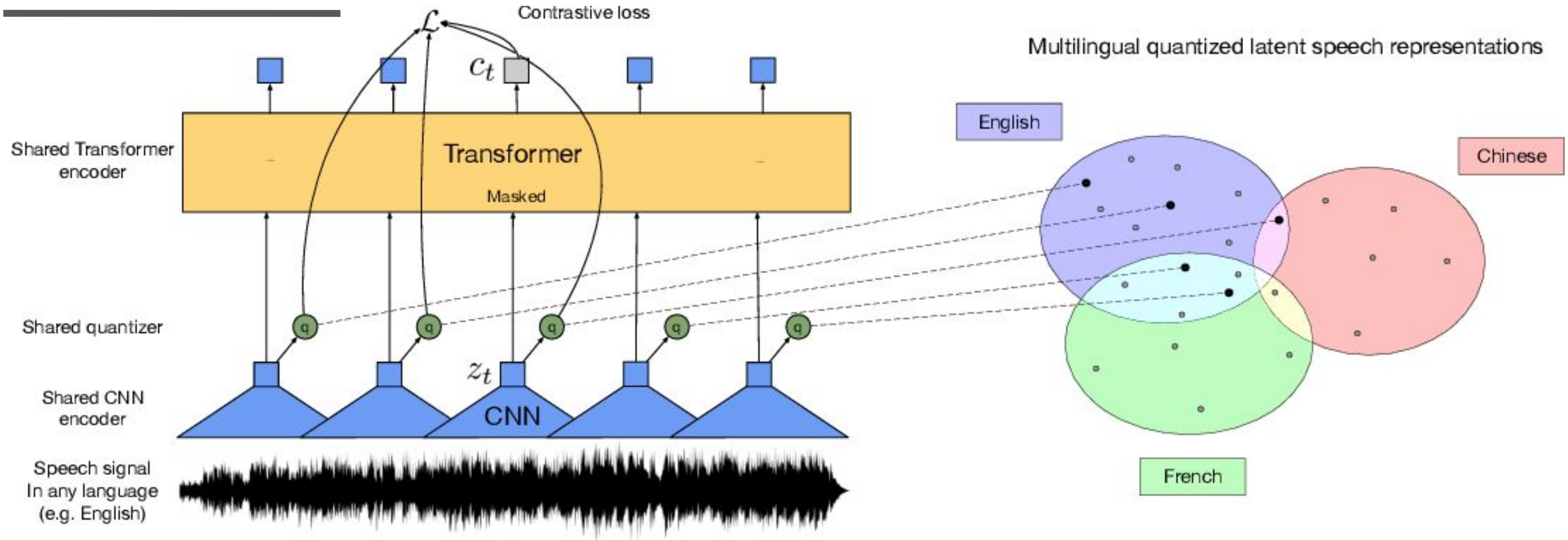
Picture courtesy: Yao, Z., Guo, L., Yang, X., Kang, W., Kuang, F., Yang, Y., ... & Povey, D. (2023, October). Zipformer: A faster and better encoder for automatic speech recognition. In The Twelfth International Conference on Learning Representations.

Whisper

Picture courtesy: https://github.com/openai/whisper



XLSR



Picture courtesy: Conneau, A., Baevski, A., Collobert, R., Mohamed, A., & Auli, M. (2020). Unsupervised cross-lingual representation learning for speech recognition. arXiv preprint arXiv:2006.13979.

Performance comparison of various acoustic models

Model	**Year**	**Test other**
GMM/HMM	–	36.4
DNN/HMM	–	12.3
TDNN	2018	7.63
Conformer	2020	3.9
Zipformer	2024	3.95
Whisper	2023	5.2
SAMBA ASR	2025	2.48

Reading materials – I

- Online Courses:
  - ASR Lectures, School of Informatics, Univ. of Edinburgh
  - CS753, IIT Bombay by Dr. Preethi Jyothi
  - Introduction to Linguistic Theory, Adam Szczegielniak, Linguistics, Psycholinguistics, Cognitive Science, University of Gdańsk
- Books:
  - Gales, M., & Young, S. (2008). The application of hidden Markov models in speech recognition. Foundations and Trends® in Signal Processing, 1(3), 195-304.
  - Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition. Online manuscript released January 12, 2025.

Reading materials – II

- Research articles:
    - Mohri et. al (2008) Speech recognition with weighted finite state transducers
    - Alex Graves, Navdeep Jaitly, "Towards End-to-End Speech Recognition with Recurrent Neural Networks",Proceedings of the 31st International Conference on Machine Learning, PMLR 32(2):1764-1772, 2014. http://proceedings.mlr.press/v32/graves14.pdf
    - R. Prabhavalkar, T. Hori, T. N. Sainath, R. Schlüter and S. Watanabe, "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024, doi: 10.1109/TASLP.2023.3328283.

Reading materials – III

- Blogs:
  - Speech Recognition — Deep Speech, CTC, Listen, Attend, and Spell
  - https://medium.com/descript/a-brief-history-of-asr-automatic-speech-recognition-b8f338d4c0e5
  - https://maelfabien.github.io/machinelearning/speech_reco/#decoding
  - Speech Recognition — GMM, HMM, by Jonathan Hui
  - Baum Welch Re-estimation: https://medium.com/analytics-vidhya/baum-welch-algorithm-for-training-a-hidden-markov-model-part-2-of-the-hmm-series-d0e393b4fb86

Demonstration

- https://espnet.github.io/espnet/installation.html
- https://medium.com/@nadirapovey/6-speech-recognition-dataset-for-asr-730f553f043d
- https://catalog.ldc.upenn.edu/LDC93S1
- Environment file: https://drive.google.com/file/d/1MT8BTFVzYNi9vaDhCqrBvenwwneFh3L-/view?usp=drive_link
- TIMIT Data: https://drive.google.com/file/d/1McZmUJUTB-aNdG6pFeHJqrPC_Zv7tL0M/view?usp=drive_link
-

TIMIT dataset

- TIMIT Dataset: 3h 8m, 20m, 9m

Thank You